SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Name: Mihiran Hanumat **PRN: 24070126113**
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Course: Data Visualization Lab **Course Code: TE7663**

Assignment Practical No. 3

Aim

To explore Power BI's data preparation and transformation capabilities using the Adventure Works dataset — covering built-in aggregations, calculated columns via Power Query, and conditional/grouping logic for deeper analysis.

Procedure Followed

Step 1 — Loading the data

- Extracted the Adventure Works Raw Data zip file to a local folder.
- Loaded the three yearly sales files (Sales Data 2020, 2021, 2022) together using Get Data → Folder → Combine Files, which stacked them into one query, later renamed Sales Data (All Years).
- Loaded the remaining 7 lookup tables individually using Get Data → Text/CSV: Customer Lookup, Product Lookup, Product Categories Lookup, Product Subcategories Lookup, Territory Lookup, Calendar Lookup, and Returns Data.
- Renamed all queries to remove the "AdventureWorks" prefix for a cleaner model.

Step 2 — Fixing a data issue

- The BirthDate column in Customer Lookup initially failed to convert to a Date type because the values were in DD/MM/YYYY format while Power Query assumed the US MM/DD/YYYY locale. Fixed this using Change Type → Using Locale → English (United Kingdom).
- A handful of rows in Customer Lookup were genuinely corrupt (blank/garbled CustomerKey and BirthDate). These were removed using Remove Errors and Remove Blank Rows so they wouldn't break later steps.

Step 3 — Building the model

- Switched to Model view and manually created relationships between the 8 base tables using their key columns (ProductKey, CustomerKey, TerritoryKey, OrderDate ↔ Calendar Date, and the Product → Subcategory → Category chain).

Step 4 — Building the working table for Part B

- Duplicated Sales Data (All Years) into a new query, Sales Analysis, and merged in columns from Product Lookup, Product Subcategories Lookup, Product Categories Lookup, Territory Lookup and Customer Lookup (using Merge Queries, matching on the respective key columns each time).
- This gave one flat table containing product, category, region and customer details alongside every sales transaction, which all the Part B and bonus calculated columns were then built on.

Step 5 — Aggregations and calculated columns

- Built the 8 aggregation measures (Part A) as DAX measures, displayed using Card visuals.
- Built all 30 calculated columns (Part B) and the 7 bonus columns/summaries (Part C) inside Power Query Editor on the Sales Analysis table, using a mix of ribbon operations (Add Column, Transform) and Custom Column formulas — listed query by query below.

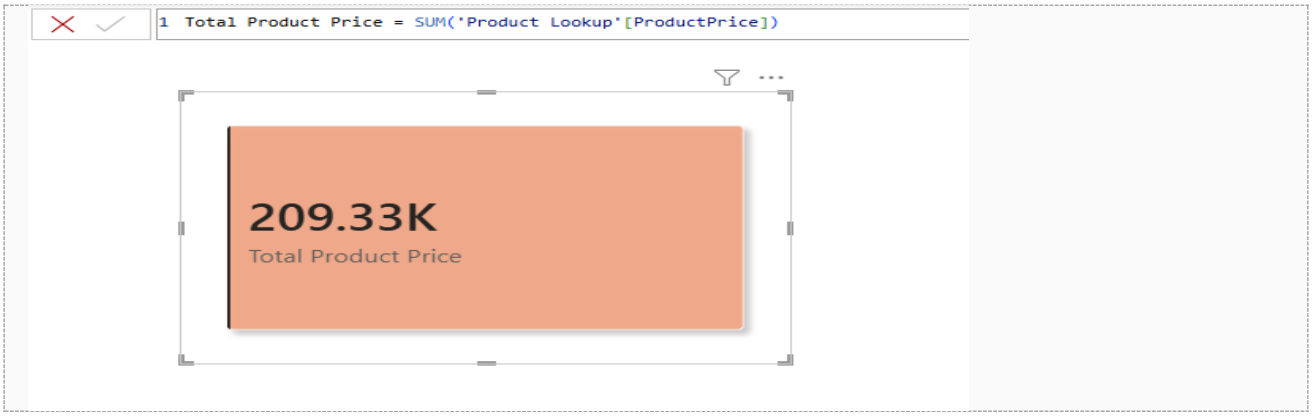
Part A: Built-in Aggregations

Each of these was created as a DAX measure (right-click table → New Measure) and then displayed on a Card visual on the report canvas.

1. SUM of ProductPrice

Adds up the price of every product to get a sense of total catalogue value and potential revenue.

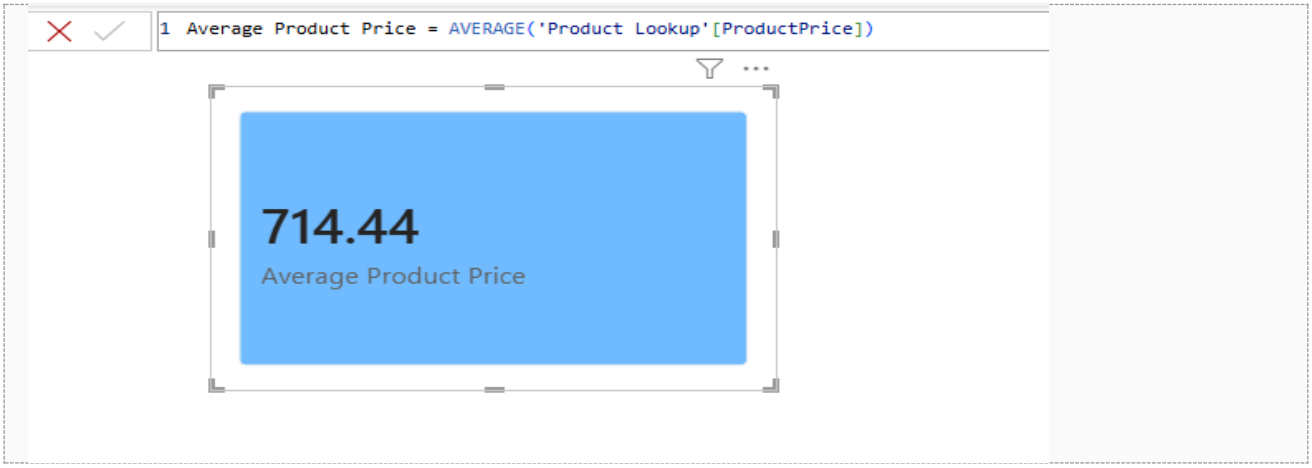
```
Total Product Price = SUM('Product Lookup'[ProductPrice])
```



2. AVERAGE of ProductPrice

Finds the typical/average selling price across all products, useful as a pricing benchmark.

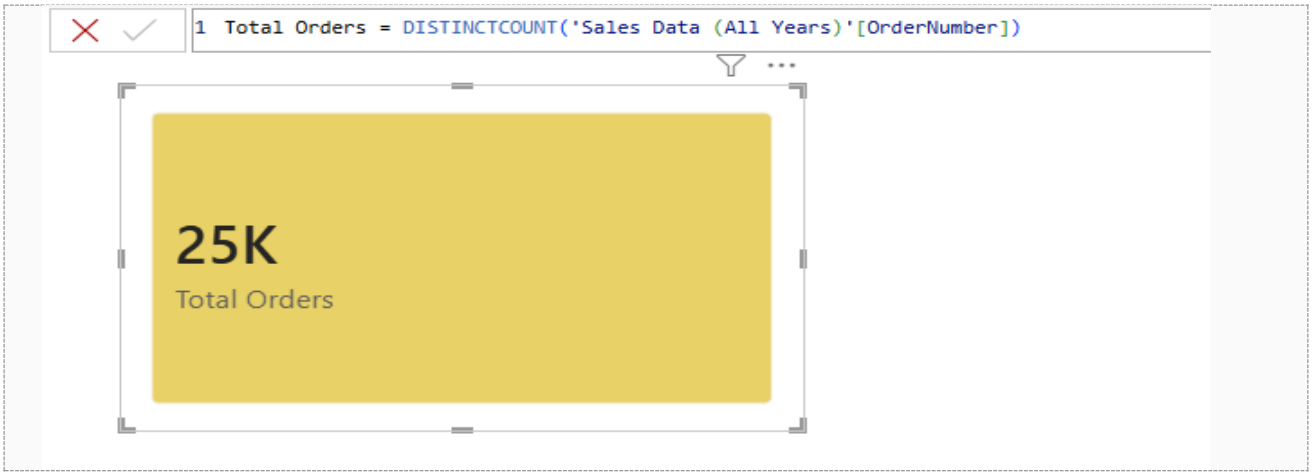
```
Average Product Price = AVERAGE('Product Lookup'[ProductPrice])
```



3. COUNT of OrderNumber (Total Orders)

Counts how many order line items exist in the data — a basic measure of transaction volume.

```
Total Orders = DISTINCTCOUNT('Sales Data (All Years)'[OrderNumber])
```



4. DISTINCTCOUNT of CustomerKey (Total Customers)

Counts unique customers only once each, even if they've placed multiple orders — this gives the true customer base size instead of an inflated number.

```
Total Customers = DISTINCTCOUNT('Sales Data (All Years)'[CustomerKey])
```



5. MIN of ProductCost

Finds the cheapest product cost, useful for spotting the lowest-cost item in the catalogue.

```
Min Product Cost = MIN('Product Lookup'[ProductCost])
```



6. MAX of ProductCost

Finds the most expensive product cost, useful for spotting outliers or premium items.

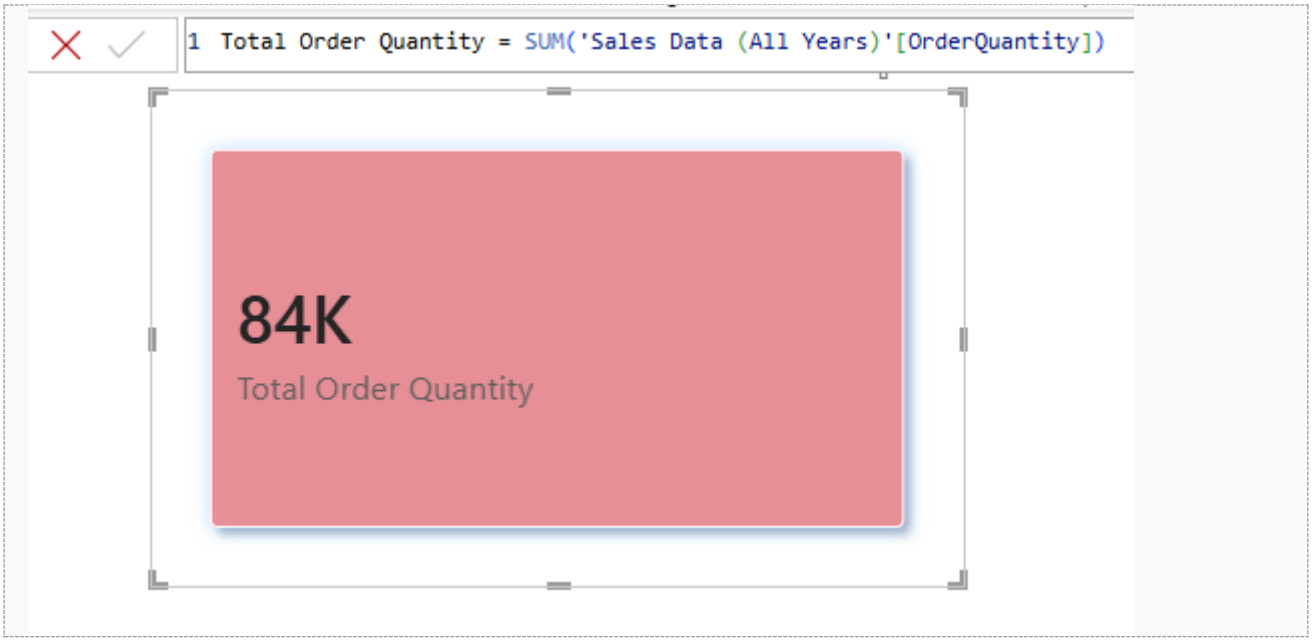
```
Max Product Cost = MAX('Product Lookup'[ProductCost])
```



7. SUM of OrderQuantity

Adds up total units sold across every order — a volume measure, separate from revenue.

```
Total Order Quantity = SUM('Sales Data (All Years)'[OrderQuantity])
```



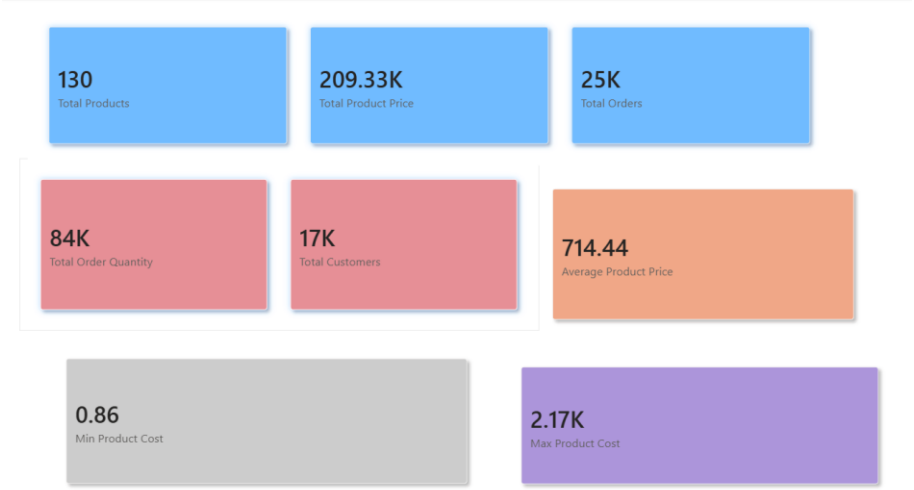
8. SUM / Count of Total Products

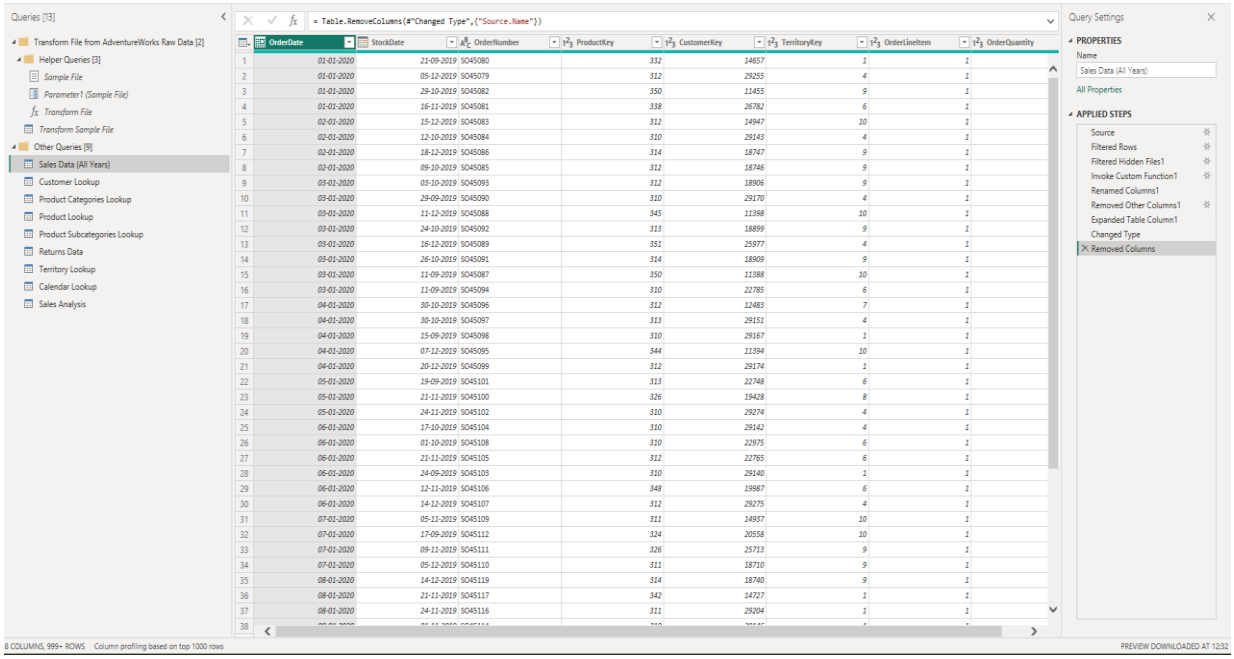
Counts how many distinct products were actually sold, as a proxy for how much of the catalogue is actively selling.

```
Total Products = DISTINCTCOUNT('Sales Data (All Years)'[ProductKey])
```



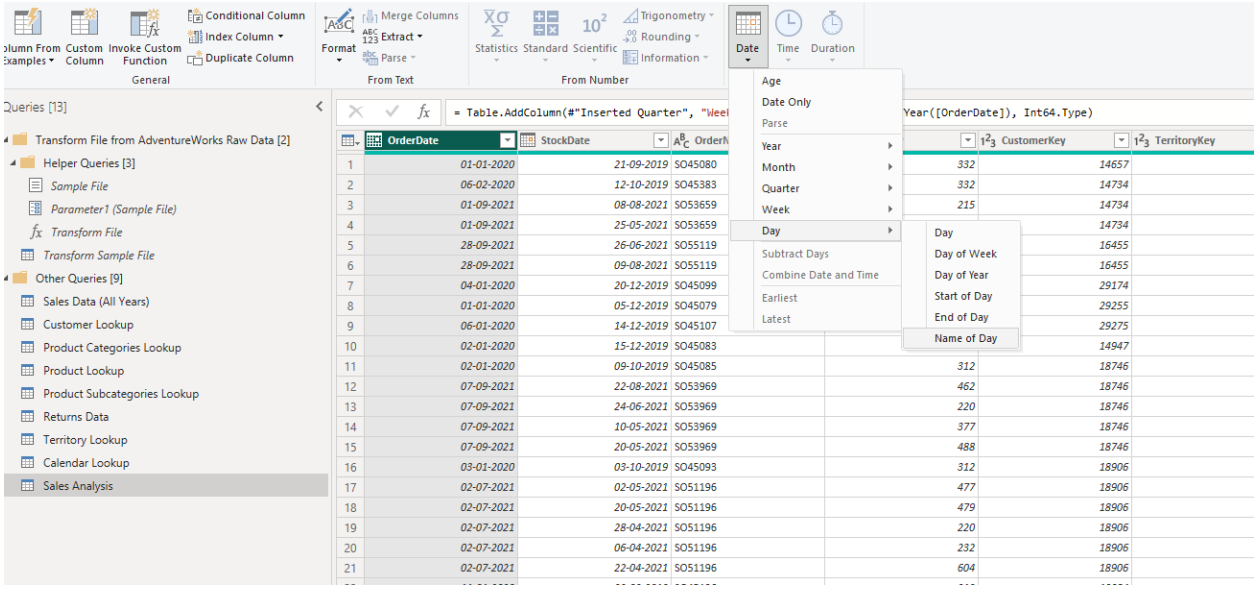
Note: Total Orders uses DISTINCTCOUNT instead of a plain COUNT, since the sales table has one row per line item, so an OrderNumber can repeat across multiple rows for the same order. Total Products uses DISTINCTCOUNT of ProductKey to mean "distinct products sold".





Part B: Calculated Columns (Power Query)

All of these columns were created inside Power Query Editor, mainly on the Sales Analysis working table (built by merging Sales Data with the Product, Category, Territory and Customer lookup tables). A Sales column (= [OrderQuantity] * [ProductPrice]) was created first, since several later tasks depend on it.



1. Extract Order Year

Pulls just the year out of a full date, so transactions can be grouped and compared year-over-year.

```
Date.Year([OrderDate])
```

2. Extract Month Name

Converts the date into a readable month label, useful for seasonal trend charts.

```
Date.MonthName([OrderDate])
```

3. Extract Quarter

Groups dates into Q1–Q4, the standard unit used in business and financial reporting.

```
Date.QuarterOfYear([OrderDate])
```

4. Extract Week Number

Gives a more granular time bucket than month, useful for short-term trend tracking.

```
Date.WeekOfYear([OrderDate])
```

5. Extract Day Name

Shows which day of the week an order was placed on, useful for spotting weekday vs weekend patterns.

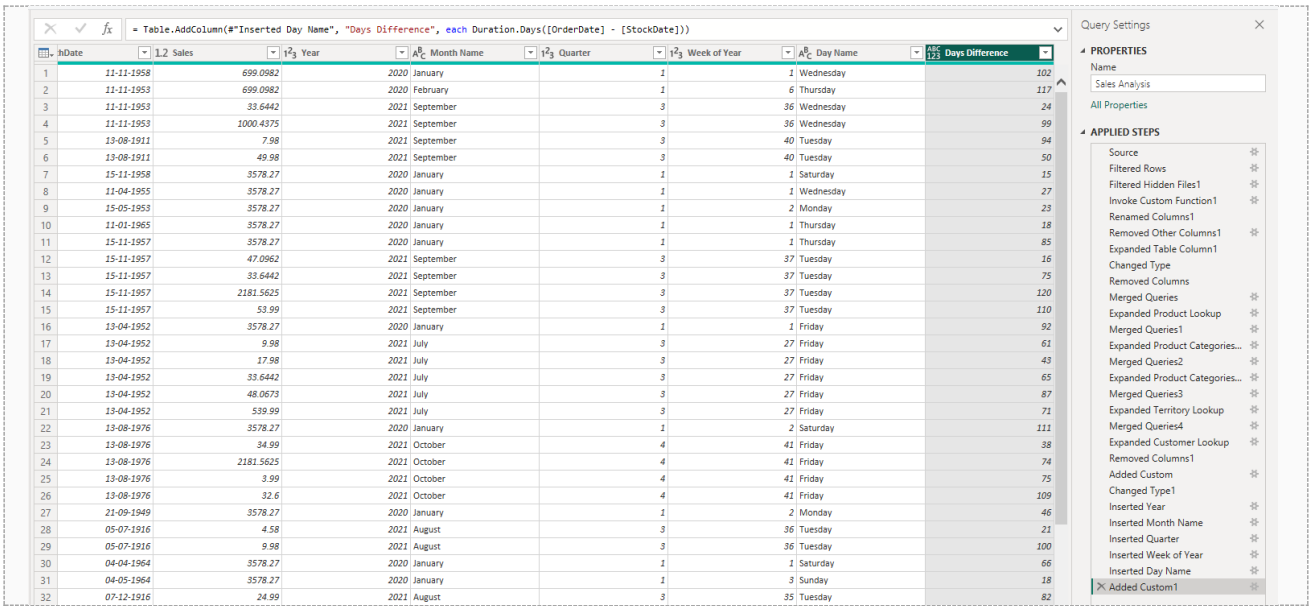
```
Date.DayOfWeekName([OrderDate])
```

Year	Month Name	Quarter	Week of Year	Day Name
2020	January	1	1	Wednesday
2020	February	1	6	Thursday
2021	September	3	36	Wednesday
2021	September	3	36	Wednesday
2021	September	3	40	Tuesday
2021	September	3	40	Tuesday
2020	January	1	1	Saturday
2020	January	1	1	Wednesday
2020	January	1	2	Monday
2020	January	1	1	Thursday
2020	January	1	1	Thursday
2021	September	3	37	Tuesday
2021	September	3	37	Tuesday
2021	September	3	37	Tuesday
2021	September	3	37	Tuesday
2020	January	1	1	Friday
2021	July	3	27	Friday
2021	July	3	27	Friday
2021	July	3	27	Friday
2021	July	3	27	Friday
2021	July	3	27	Friday
2020	January	1	2	Saturday
2021	October	4	41	Friday
2021	October	4	41	Friday
2021	October	4	41	Friday
2021	October	4	41	Friday
2020	January	1	2	Monday
2021	August	3	36	Tuesday
2021	August	3	36	Tuesday
2020	January	1	1	Saturday
2020	January	1	3	Sunday
2021	August	3	35	Tuesday
2020	January	1	3	Sunday
2020	January	1	3	Monday
2020	January	1	3	Tuesday
2021	July	3	31	Sunday
2021	July	3	31	Sunday

6. Calculate Days Difference

Measures the gap in days between two dates — here, how long between the order date and stock date, which can indicate dispatch or inventory lag.

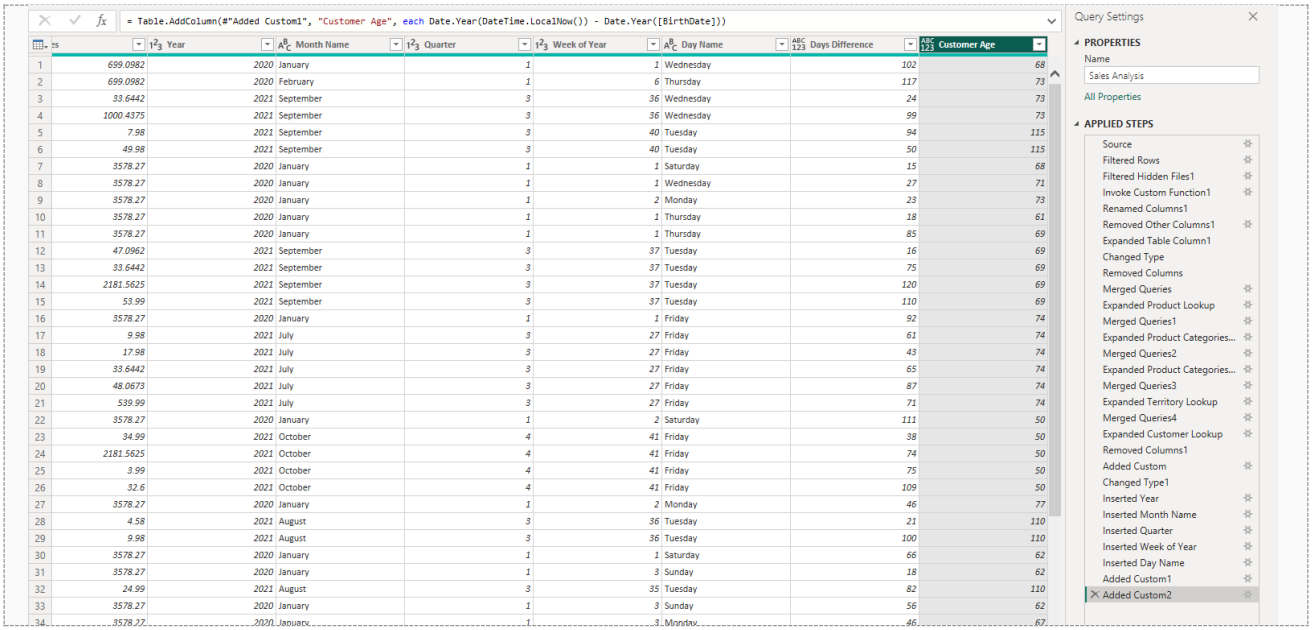
```
Duration.Days([OrderDate] - [StockDate])
```



7. Calculate Customer Age

Derives a demographic attribute (age) from birth date, useful for segmenting customers by age group.

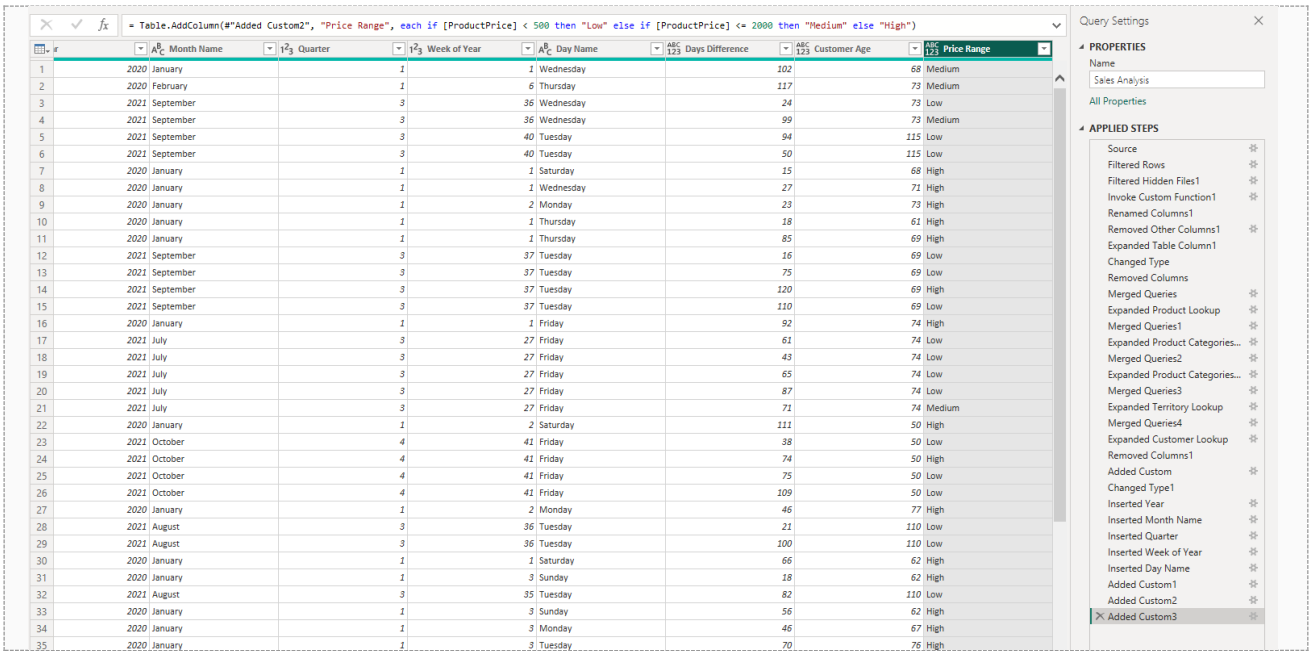
```
Date.Year(DateTime.LocalNow()) - Date.Year([BirthDate])
```



8. Create Price Range (Low/Medium/High)

Buckets continuous price values into simple tiers, which are much easier to filter and visualize than raw numbers.

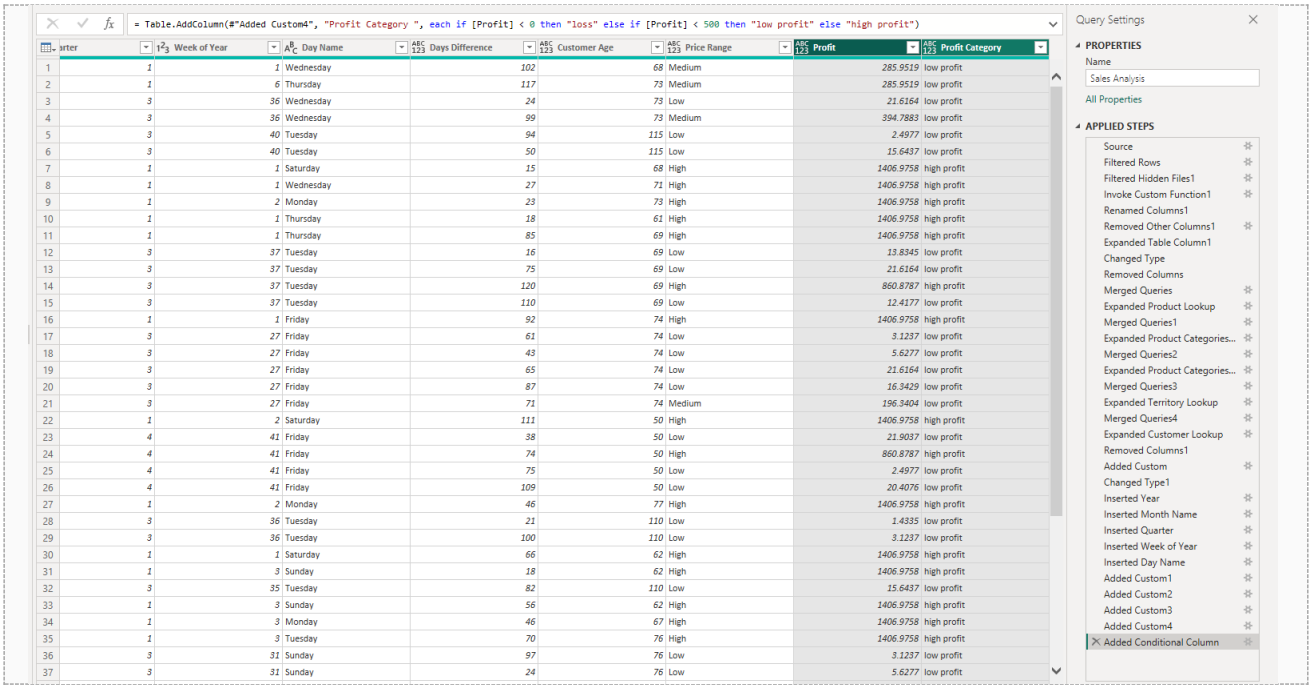
```
if [ProductPrice] < 500 then "Low" else if [ProductPrice] <= 2000 then "Medium" else "High"
```



9. Create Profit Category (Loss/Low/High Profit)

Flags whether a product is sold at a loss or at a low/high profit — directly useful for profitability analysis.

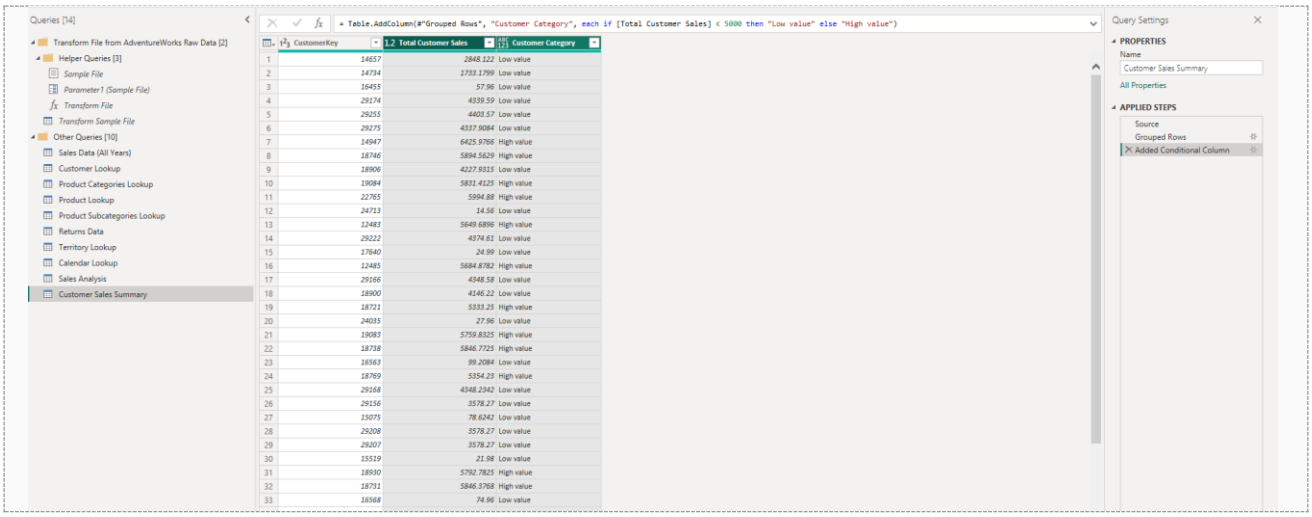
```
Profit = [ProductPrice] - [ProductCost]
if [Profit] < 0 then "Loss" else if [Profit] < 500 then "Low Profit" else "High Profit"
```



10. Categorize Customers based on Sales

Segments customers into value tiers based on how much they've spent in total — a similar idea to customer-value segmentation used in marketing.

```
Table.Group(Sales Analysis, {"CustomerKey"}, {{"Total Customer Sales", each
List.Sum([Sales]), type number}})
then: if [Total Customer Sales] < 5000 then "Low Value" else "High Value"
```



11. Round Price Amount

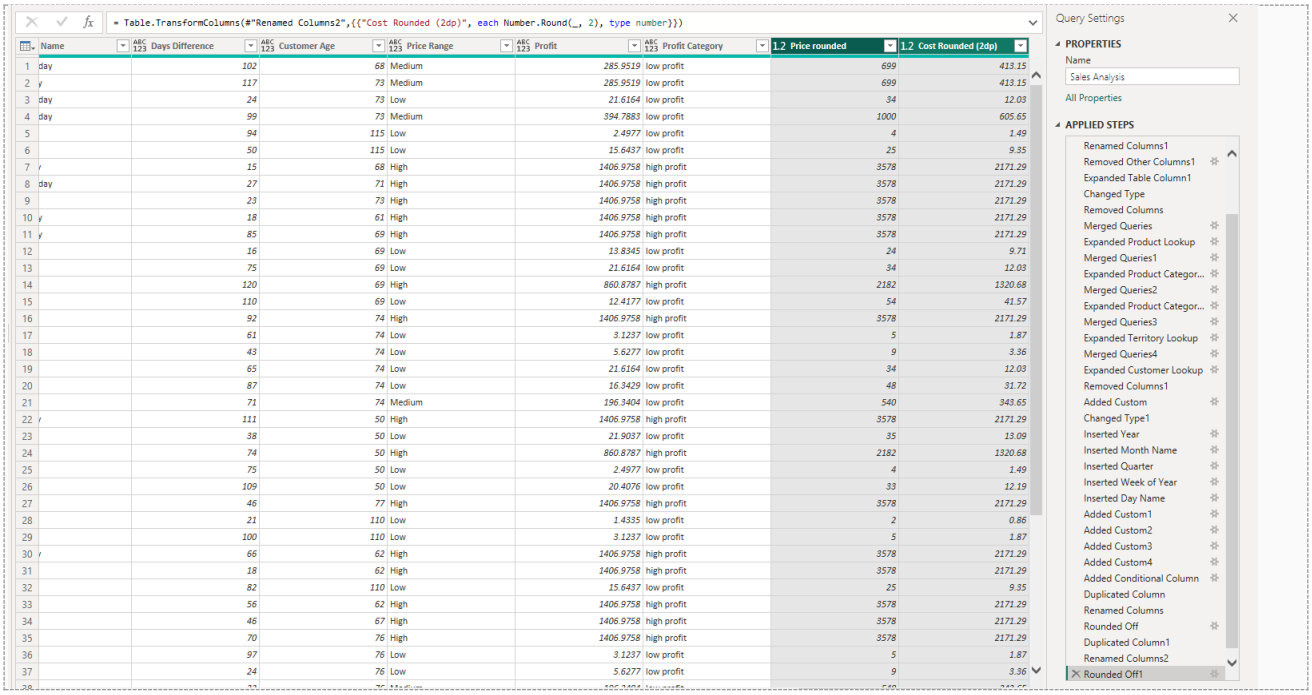
Removes decimal clutter so prices display more cleanly in reports and visuals.

```
Number.Round([ProductPrice], 0)
```

12. Round Cost Amount (2 decimals)

Standardizes cost values to the usual 2-decimal currency format.

```
Number.Round([ProductCost], 2)
```



13. Convert Product Name to UPPERCASE

Standardizes text casing, useful for consistent matching, searching, or for display in headers/labels.

```
Text.Upper([ProductName])
```

14. Convert Product Name to lowercase

Same idea in reverse — standardizes casing, useful for case-insensitive comparisons.

```
Text.Lower([ProductName])
```

15. Trim Extra Spaces

Removes stray leading/trailing spaces that often cause silent mismatches when filtering or joining data.

```
Text.Trim([ProductName])
```

16. Clean Product Name

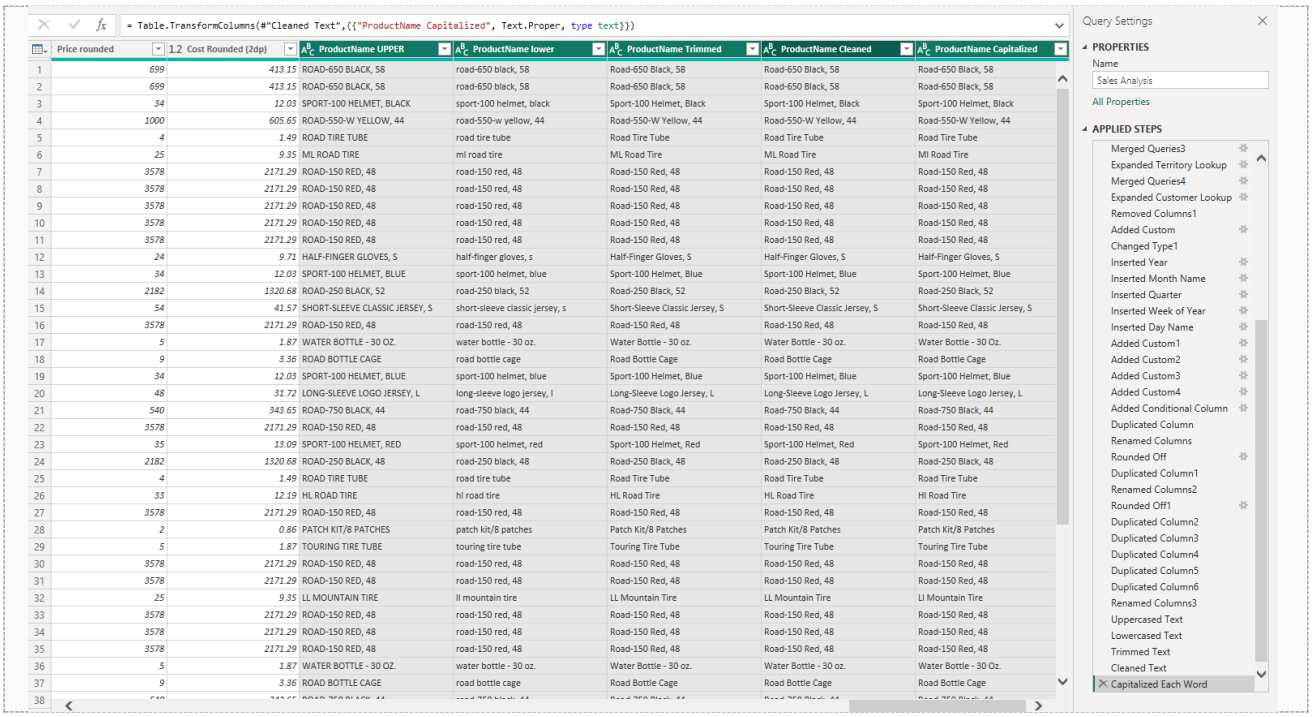
Strips out non-printable characters that sometimes creep in from exports or copy-pasted data.

```
Text.Clean([ProductName])
```

17. Capitalize Each Word

Formats text into Proper Case, which reads better in reports than all-caps or all-lowercase text.

```
Text.Proper([ProductName])
```

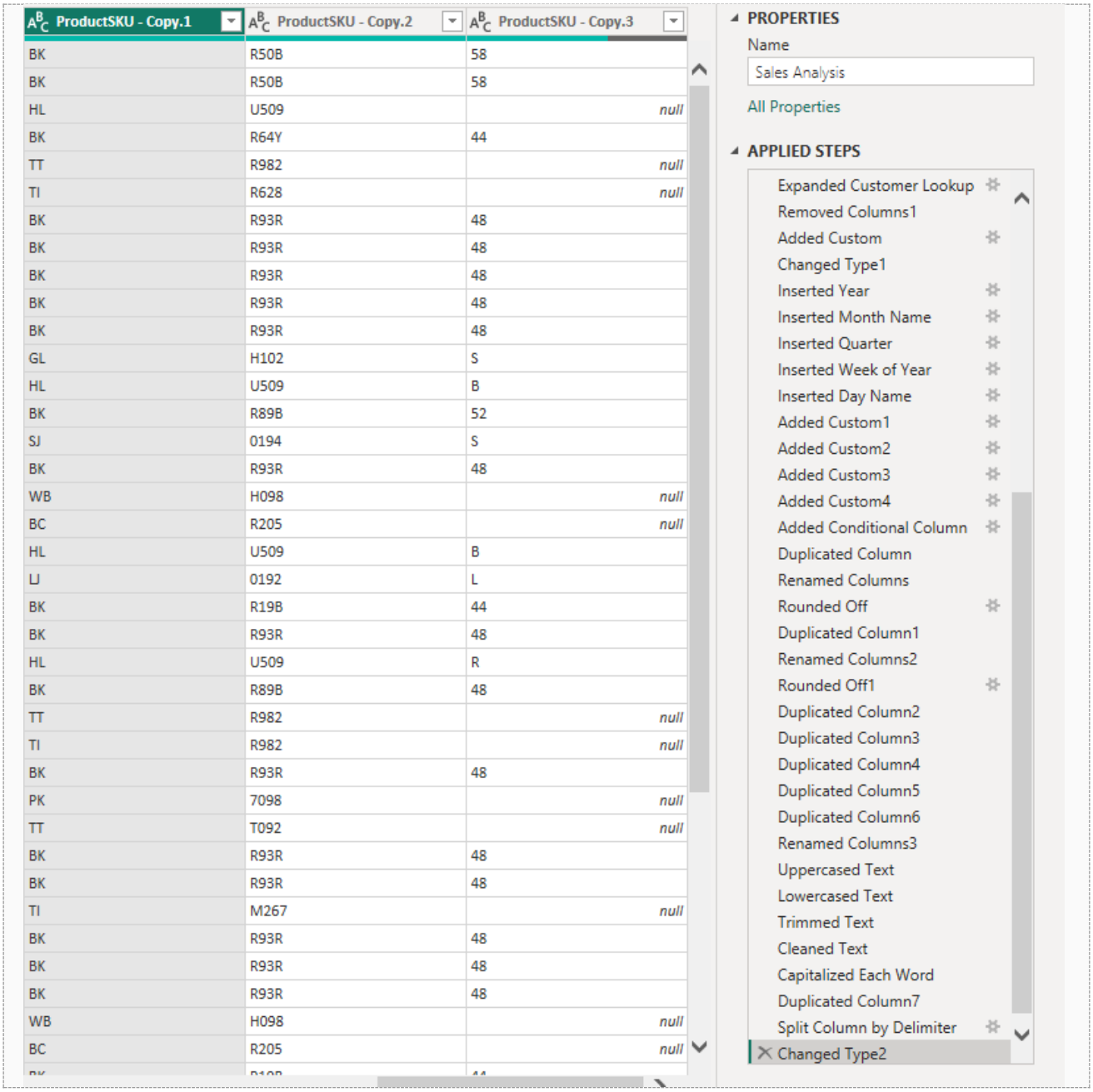


	Price rounded	Cost Rounded (2dp)	ProductName UPPER	ProductName lower	ProductName Trimmed	ProductName Cleaned	ProductName Capitalized
1	699	413.15	ROAD-650 BLACK, 58	road-650 black, 58	Road-650 Black, 58	Road-650 Black, 58	Road-650 Black, 58
2	699	413.15	ROAD-650 BLACK, 58	road-650 black, 58	Road-650 Black, 58	Road-650 Black, 58	Road-650 Black, 58
3	34	12.03	SPORT-100 HELMET, BLACK	sport-100 helmet, black	Sport-100 Helmet, Black	Sport-100 Helmet, Black	Sport-100 Helmet, Black
4	1000	605.65	ROAD-550-W YELLOW, 44	road-550-w yellow, 44	Road-550-W Yellow, 44	Road-550-W Yellow, 44	Road-550-W Yellow, 44
5	4	1.49	ROAD TIRE TUBE	road tire tube	Road Tire Tube	Road Tire Tube	Road Tire Tube
6	25	9.35	ML ROAD TIRE	ml road tire	ML Road Tire	ML Road Tire	Ml Road Tire
7	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
8	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
9	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
10	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
11	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
12	24	9.71	HALF-FINGER GLOVES, S	half-finger gloves, s	Half-Finger Gloves, S	Half-Finger Gloves, S	Half-Finger Gloves, S
13	34	12.03	SPORT-100 HELMET, BLUE	sport-100 helmet, blue	Sport-100 Helmet, Blue	Sport-100 Helmet, Blue	Sport-100 Helmet, Blue
14	2182	1320.68	ROAD-250 BLACK, 52	road-250 black, 52	Road-250 Black, 52	Road-250 Black, 52	Road-250 Black, 52
15	54	41.57	SHORT-SLEEVE CLASSIC JERSEY, S	short-sleeve classic jersey, s	Short-Sleeve Classic Jersey, S	Short-Sleeve Classic Jersey, S	Short-Sleeve Classic Jersey, S
16	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
17	5	1.87	WATER BOTTLE - 30 OZ.	water bottle - 30 oz.	Water Bottle - 30 oz.	Water Bottle - 30 oz.	Water Bottle - 30 Oz.
18	9	3.36	ROAD BOTTLE CAGE	road bottle cage	Road Bottle Cage	Road Bottle Cage	Road Bottle Cage
19	34	12.03	SPORT-100 HELMET, BLUE	sport-100 helmet, blue	Sport-100 Helmet, Blue	Sport-100 Helmet, Blue	Sport-100 Helmet, Blue
20	48	31.72	LONG-SLEEVE LOGO JERSEY, L	long-sleeve logo jersey, l	Long-Sleeve Logo Jersey, L	Long-Sleeve Logo Jersey, L	Long-Sleeve Logo Jersey, L
21	540	343.65	ROAD-750 BLACK, 44	road-750 black, 44	Road-750 Black, 44	Road-750 Black, 44	Road-750 Black, 44
22	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
23	35	13.09	SPORT-100 HELMET, RED	sport-100 helmet, red	Sport-100 Helmet, Red	Sport-100 Helmet, Red	Sport-100 Helmet, Red
24	2182	1320.68	ROAD-250 BLACK, 48	road-250 black, 48	Road-250 Black, 48	Road-250 Black, 48	Road-250 Black, 48
25	4	1.49	ROAD TIRE TUBE	road tire tube	Road Tire Tube	Road Tire Tube	Road Tire Tube
26	33	12.19	HL ROAD TIRE	hl road tire	HL Road Tire	HL Road Tire	Hl Road Tire
27	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
28	2	0.86	PATCH KIT/8 PATCHES	patch kit/8 patches	Patch Kit/8 Patches	Patch Kit/8 Patches	Patch Kit/8 Patches
29	5	1.87	TOURING TIRE TUBE	touring tire tube	Touring Tire Tube	Touring Tire Tube	Touring Tire Tube
30	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
31	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
32	25	9.35	LL MOUNTAIN TIRE	ll mountain tire	LL Mountain Tire	LL Mountain Tire	Ll Mountain Tire
33	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
34	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
35	3578	2171.29	ROAD-150 RED, 48	road-150 red, 48	Road-150 Red, 48	Road-150 Red, 48	Road-150 Red, 48
36	5	1.87	WATER BOTTLE - 30 OZ.	water bottle - 30 oz.	Water Bottle - 30 oz.	Water Bottle - 30 oz.	Water Bottle - 30 Oz.
37	9	3.36	ROAD BOTTLE CAGE	road bottle cage	Road Bottle Cage	Road Bottle Cage	Road Bottle Cage
38	540	343.65	ROAD-750 BLACK, 44	road-750 black, 44	Road-750 Black, 44	Road-750 Black, 44	Road-750 Black, 44

18. Split Product Name (SKU) by "-"

Breaks a compound code into its separate parts, so each part can be analysed or filtered individually.

```
Table.SplitColumn(Sales Analysis, "ProductSKU", Splitter.SplitTextByDelimiter("-"))
```



ProductSKU - Copy.1	ProductSKU - Copy.2	ProductSKU - Copy.3
BK	R50B	58
BK	R50B	58
HL	U509	null
BK	R64Y	44
TT	R982	null
TI	R628	null
BK	R93R	48
BK	R93R	48
BK	R93R	48
BK	R93R	48
BK	R93R	48
GL	H102	S
HL	U509	B
BK	R89B	52
SJ	0194	S
BK	R93R	48
WB	H098	null
BC	R205	null
HL	U509	B
LJ	0192	L
BK	R19B	44
BK	R93R	48
HL	U509	R
BK	R89B	48
TT	R982	null
TI	R982	null
BK	R93R	48
PK	7098	null
TT	T092	null
BK	R93R	48
BK	R93R	48
TI	M267	null
BK	R93R	48
BK	R93R	48
BK	R93R	48
WB	H098	null
BC	R205	null

PROPERTIES

Name

Sales Analysis

All Properties

APPLIED STEPS

- Expanded Customer Lookup
- Removed Columns1
- Added Custom
- Changed Type1
- Inserted Year
- Inserted Month Name
- Inserted Quarter
- Inserted Week of Year
- Inserted Day Name
- Added Custom1
- Added Custom2
- Added Custom3
- Added Custom4
- Added Conditional Column
- Duplicated Column
- Renamed Columns
- Rounded Off
- Duplicated Column1
- Renamed Columns2
- Rounded Off1
- Duplicated Column2
- Duplicated Column3
- Duplicated Column4
- Duplicated Column5
- Duplicated Column6
- Renamed Columns3
- Uppercased Text
- Lowercased Text
- Trimmed Text
- Cleaned Text
- Capitalized Each Word
- Duplicated Column7
- Split Column by Delimiter
- Changed Type2

19. Extract First 8 Characters

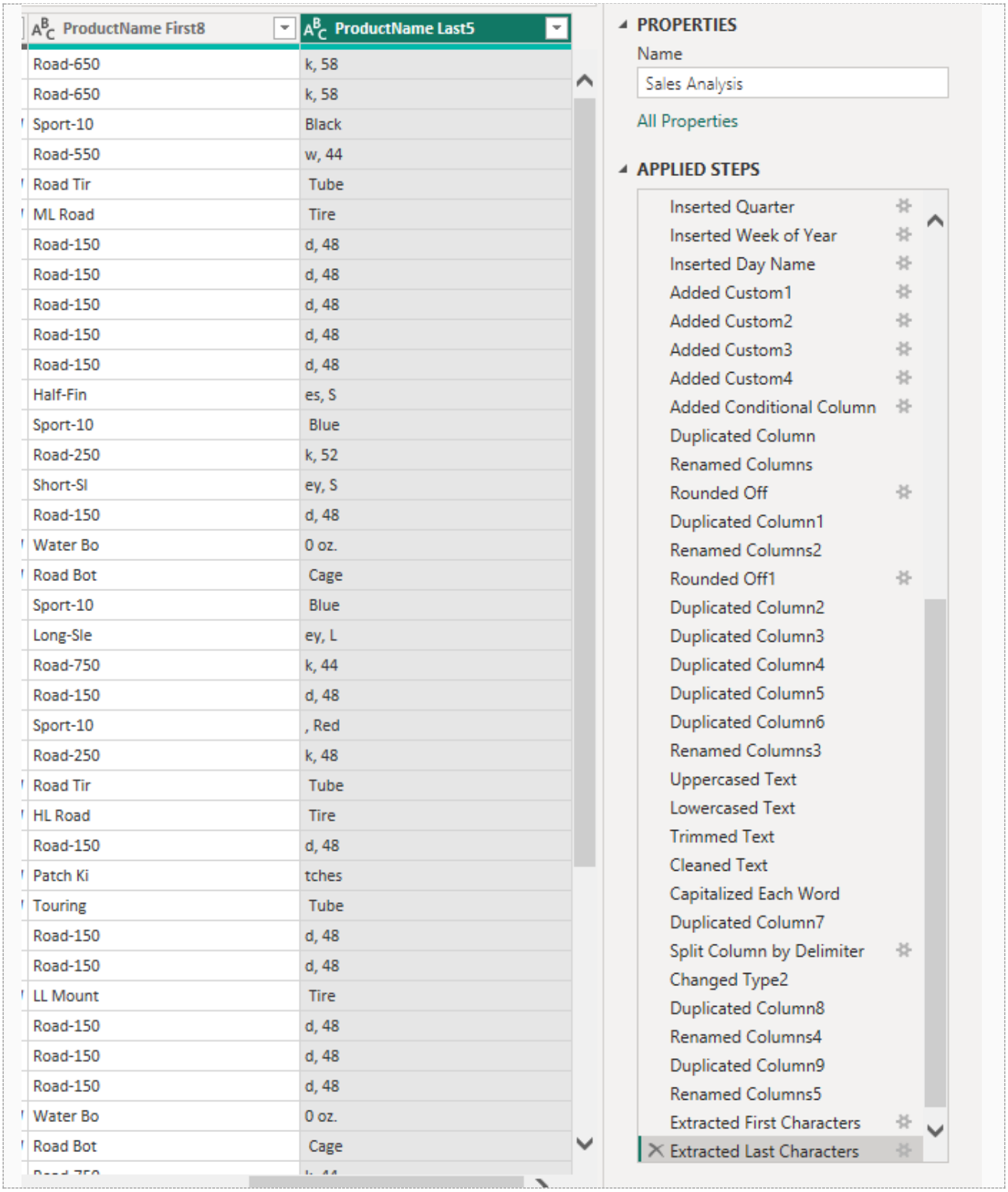
Pulls out a fixed-length prefix from text — useful for generating short codes or IDs from longer names.

```
Text.Start([ProductName], 8)
```

20. Extract Last 5 Characters

Pulls out a fixed-length suffix — useful when a naming convention stores information (like size or version) at the end of the text.

```
Text.End([ProductName], 5)
```

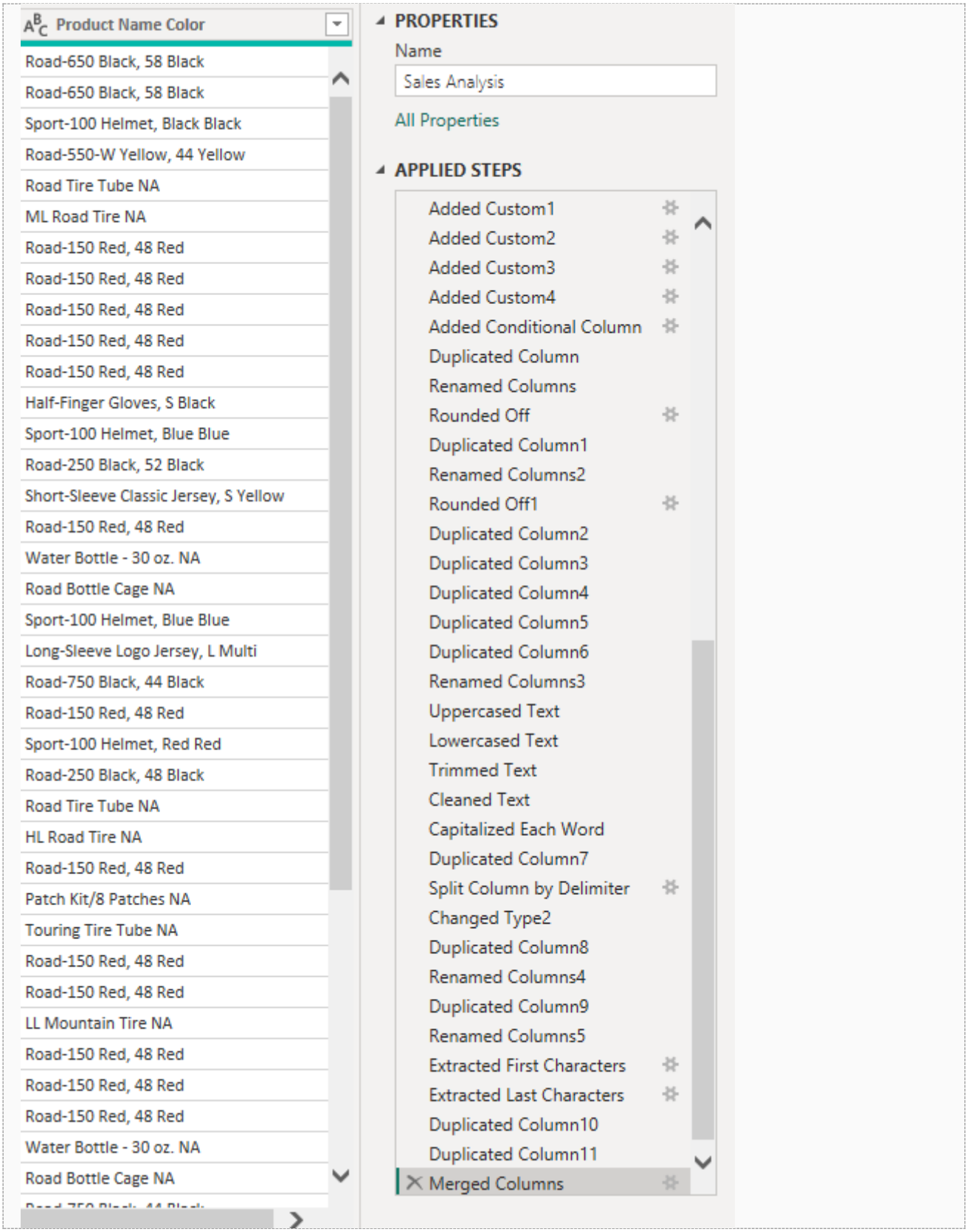


ProductName First8	ProductName Last5
Road-650	k, 58
Road-650	k, 58
Sport-10	Black
Road-550	w, 44
Road Tir	Tube
ML Road	Tire
Road-150	d, 48
Road-150	d, 48
Road-150	d, 48
Road-150	d, 48
Road-150	d, 48
Half-Fin	es, S
Sport-10	Blue
Road-250	k, 52
Short-Sl	ey, S
Road-150	d, 48
Water Bo	0 oz.
Road Bot	Cage
Sport-10	Blue
Long-Sle	ey, L
Road-750	k, 44
Road-150	d, 48
Sport-10	, Red
Road-250	k, 48
Road Tir	Tube
HL Road	Tire
Road-150	d, 48
Patch Ki	tches
Touring	Tube
Road-150	d, 48
Road-150	d, 48
LL Mount	Tire
Road-150	d, 48
Road-150	d, 48
Road-150	d, 48
Water Bo	0 oz.
Road Bot	Cage

21. Merge Product Name and Color

Combines two attributes into a single readable label, handy for a friendlier display column.

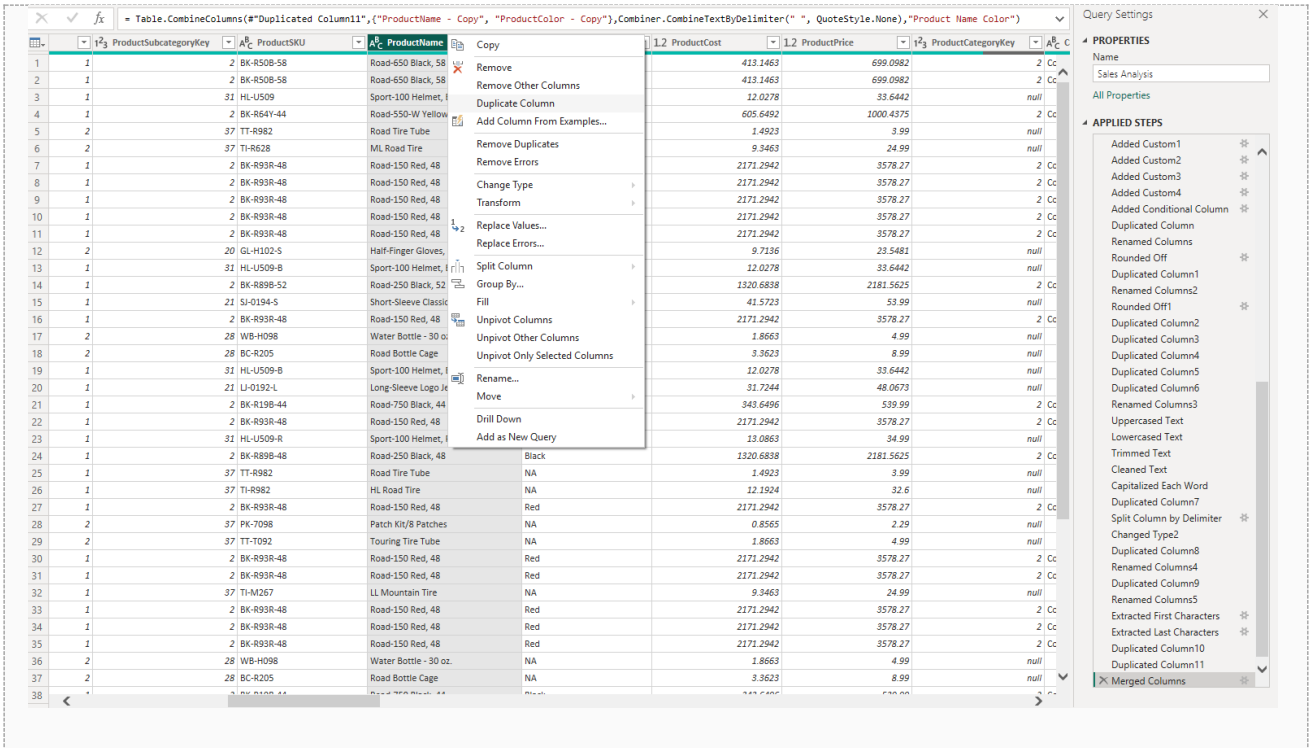
```
[ProductName] & " " & [ProductColor]
```

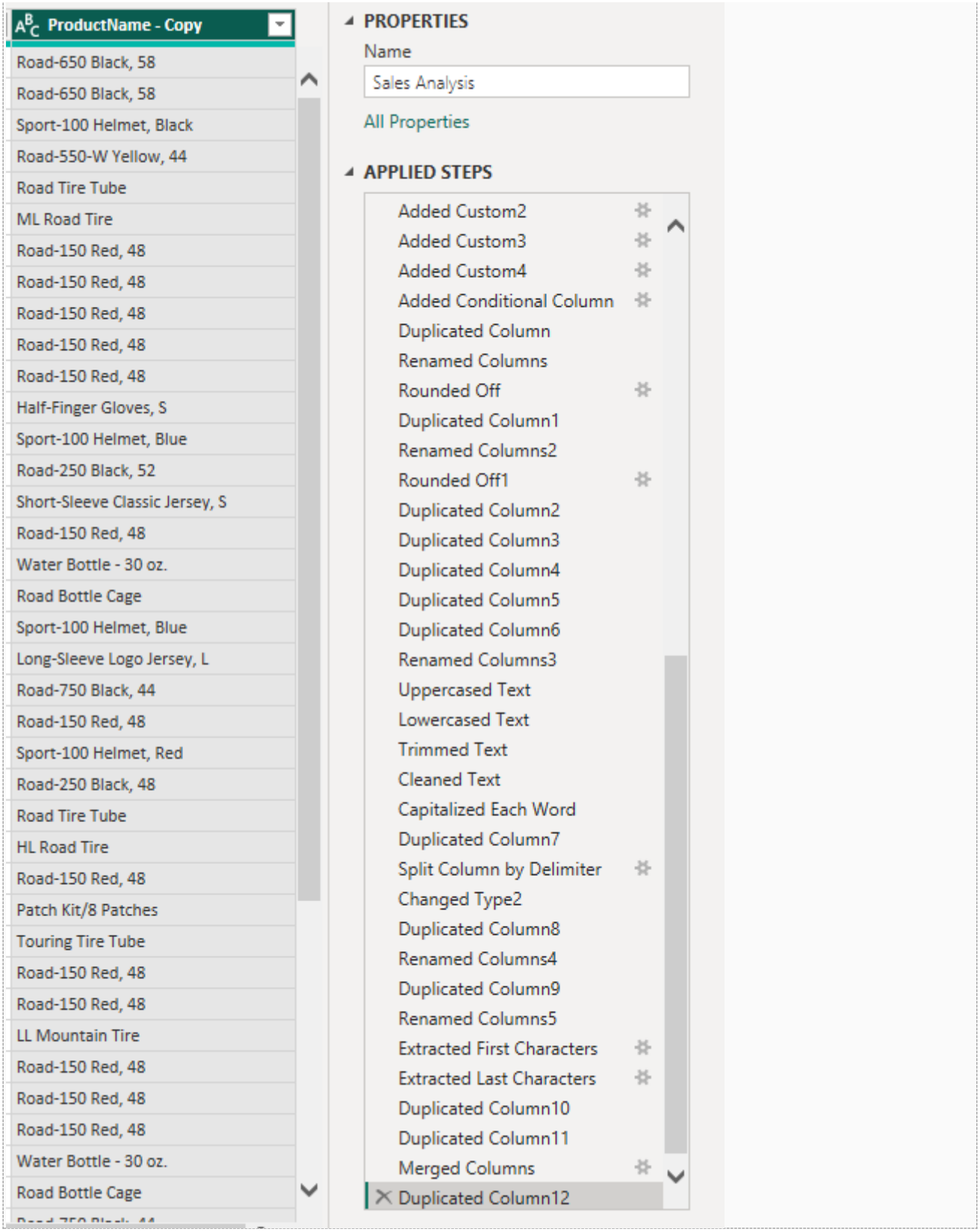


22. Duplicate Product Name Column

Keeps the original column untouched while giving a copy to experiment or transform on — a good safety-net habit in Power Query.

```
Table.DuplicateColumn(Sales Analysis, "ProductName", "ProductName - Copy")
```

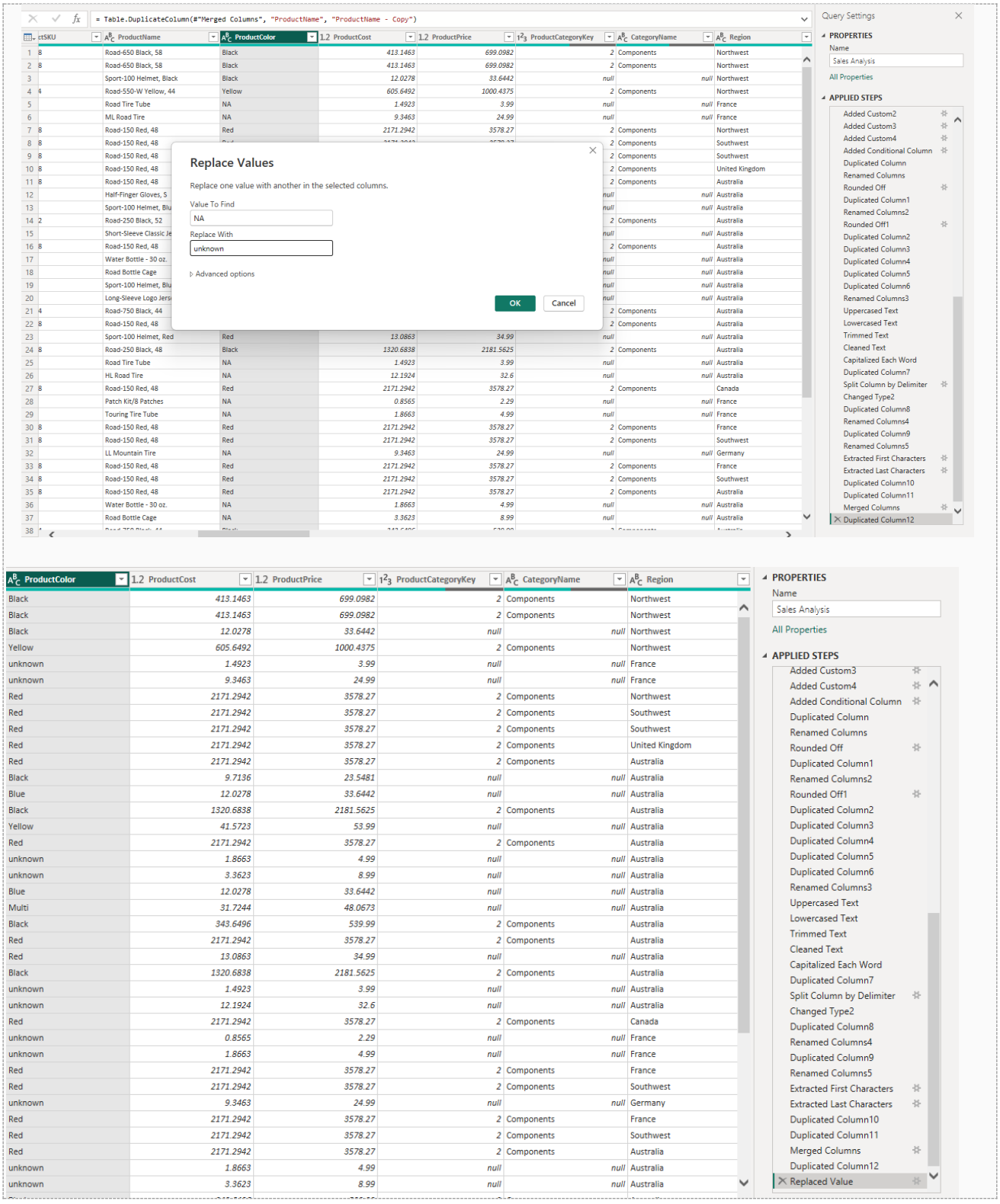

= Table.CombineColumns(#"Duplicated Column11",{"ProductName - Copy", "ProductColor - Copy"},Combiner.CombineTextByDelimiter(" ", QuoteStyle.None),"Product Name Color")
ProductSubcategoryKey
ProductSKU
ProductName
ProductCost
ProductPrice
ProductCategoryKey
Copy
Remove
Remove Other Columns
Duplicate Column
Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
Replace Values...
Replace Errors...
Split Column
Group By...
Fill
Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
Rename...
Move
Drill Down
Add as New Query
Query Settings
PROPERTIES
Name
Sales Analysis
All Properties
APPLIED STEPS
Added Custom1
Added Custom2
Added Custom3
Added Custom4
Added Conditional Column
Duplicated Column
Renamed Columns
Rounded Off
Duplicated Column1
Renamed Columns2
Rounded Off1
Duplicated Column2
Duplicated Column3
Duplicated Column4
Duplicated Column5
Duplicated Column6
Renamed Columns3
Uppercased Text
Lowercased Text
Trimmed Text
Cleaned Text
Capitalized Each Word
Duplicated Column7
Split Column by Delimiter
Changed Type2
Duplicated Column8
Renamed Columns4
Duplicated Column9
Renamed Columns5
Extracted First Characters
Extracted Last Characters
Duplicated Column10
Duplicated Column11
Merged Columns



23. Replace Missing Values

Fills blank/NA entries with a placeholder so they don't silently break later calculations or visuals.

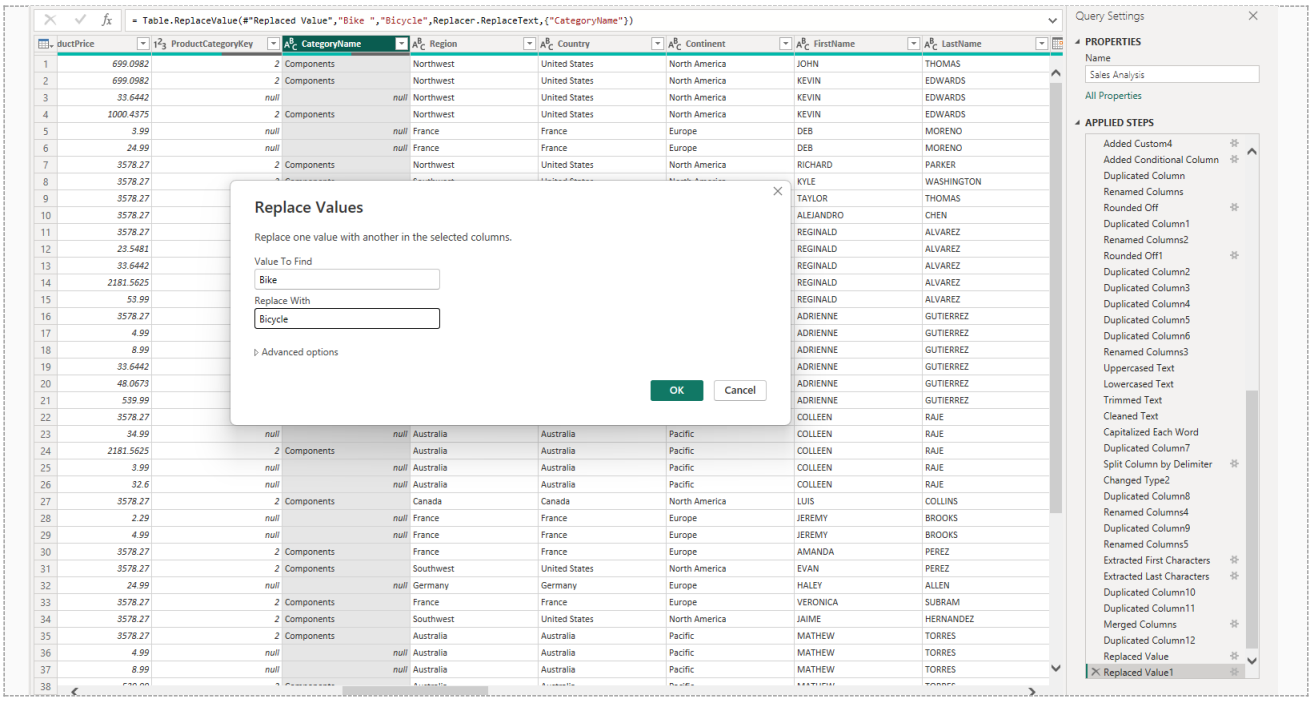
```
Table.ReplaceValue(Sales Analysis, "NA", "Unknown", Replacer.ReplaceValue,
{"ProductColor"})
```



24. Replace Bike with Bicycle

Demonstrates simple find-and-replace, useful for standardizing inconsistent terminology in text data.

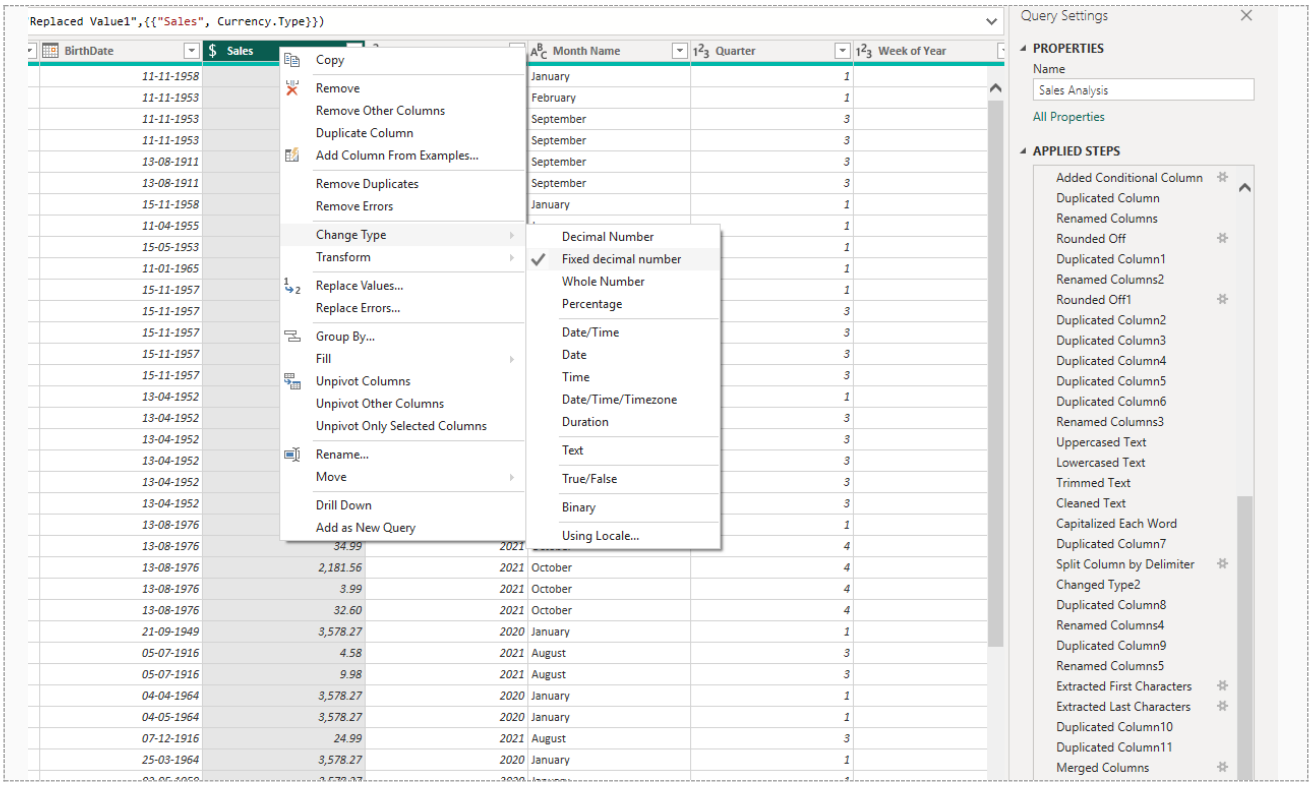
```
Table.ReplaceValue(Sales Analysis, "Bike", "Bicycle", Replacer.ReplaceText,
{"ProductName"})
```



25. Change Sales Data Type to Fixed Decimal

Forces the column to store numbers with consistent precision, which matters for accurate downstream sums and averages.

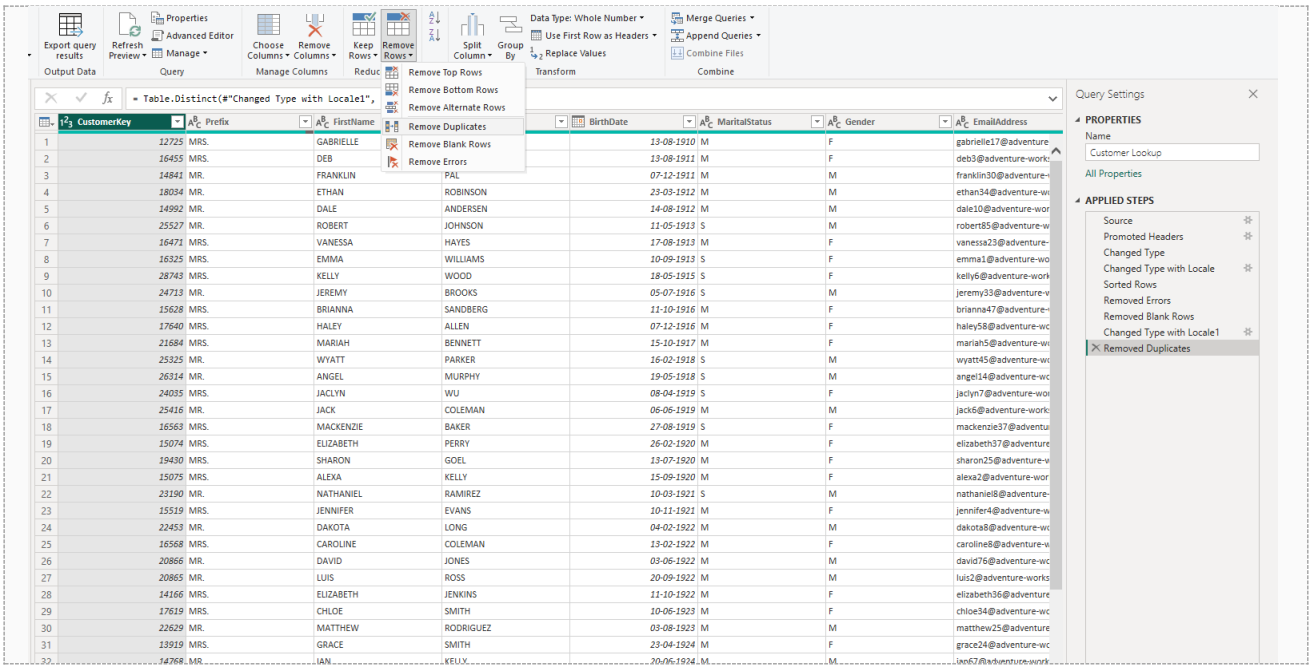
```
Table.TransformColumnTypes(Sales Analysis, {{"Sales", type number}})
```



26. Remove Duplicate Customers

Eliminates repeated key values so customer counts and lookups stay accurate.

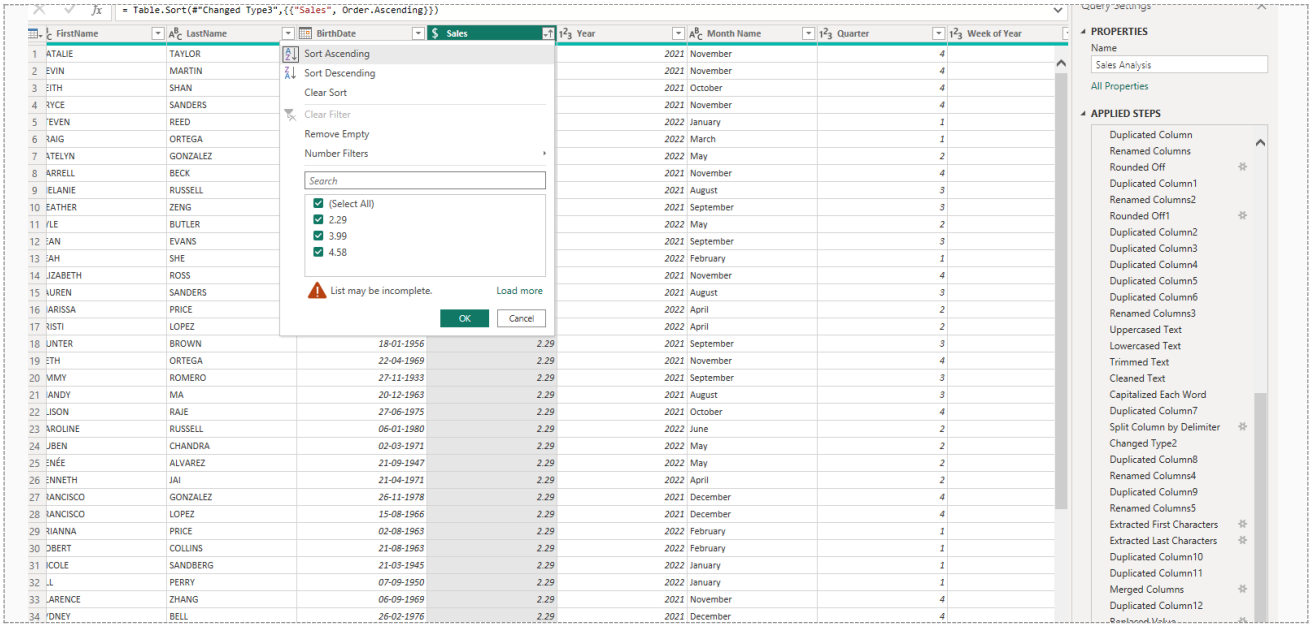
```
Table.Distinct(Customer Lookup, {"CustomerKey"})
```



27. Sort Products by Sales Amount

Reorders rows so the highest (or lowest) performing transactions are immediately visible.

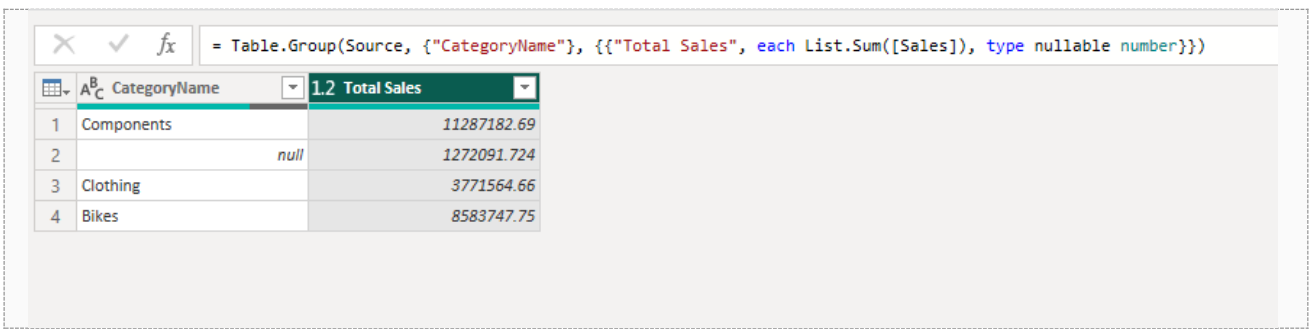
```
Table.Sort(Sales Analysis, {{"Sales", Order.Descending}})
```



28. Group by Product Category and Sum Sales

Rolls up transaction-level rows into one row per category, showing total revenue per category.

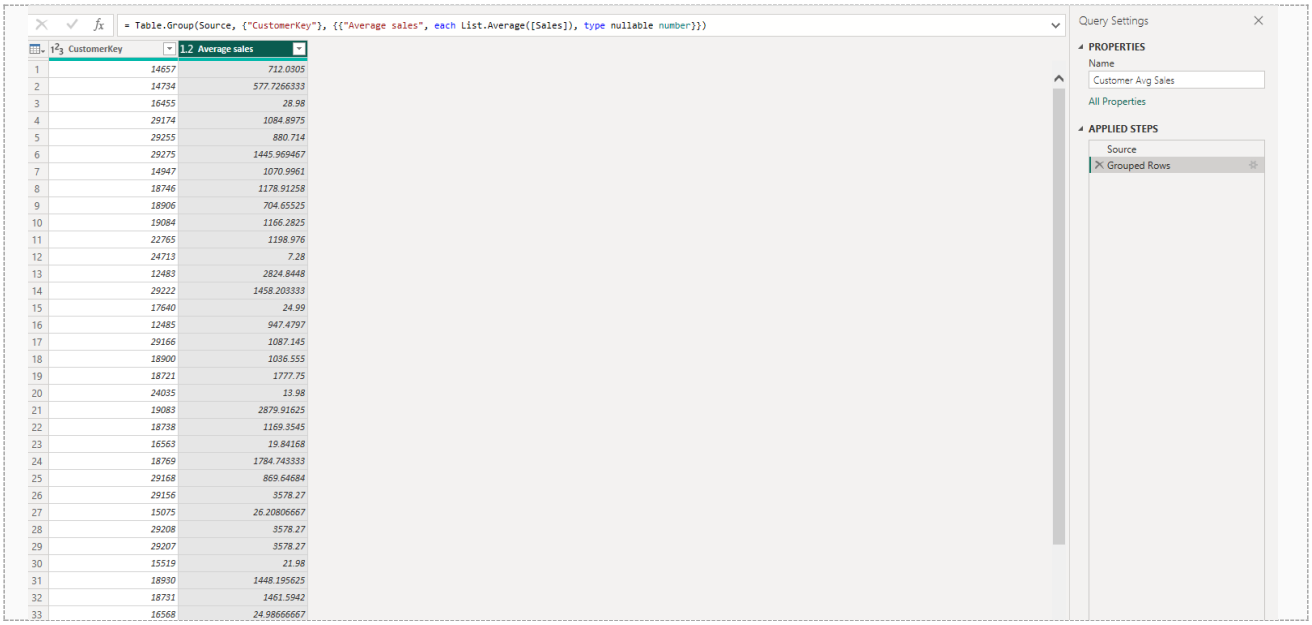
```
Table.Group(Sales Analysis, {"CategoryName"}, {{"Total Sales", each List.Sum([Sales]), type number}})
```



29. Group by Customer and Average Sales

Rolls up rows into one row per customer, showing average spend — useful for understanding typical customer value.

```
Table.Group(Sales Analysis, {"CustomerKey"}, {{"Average Sales", each List.Average([Sales]), type number}})
```

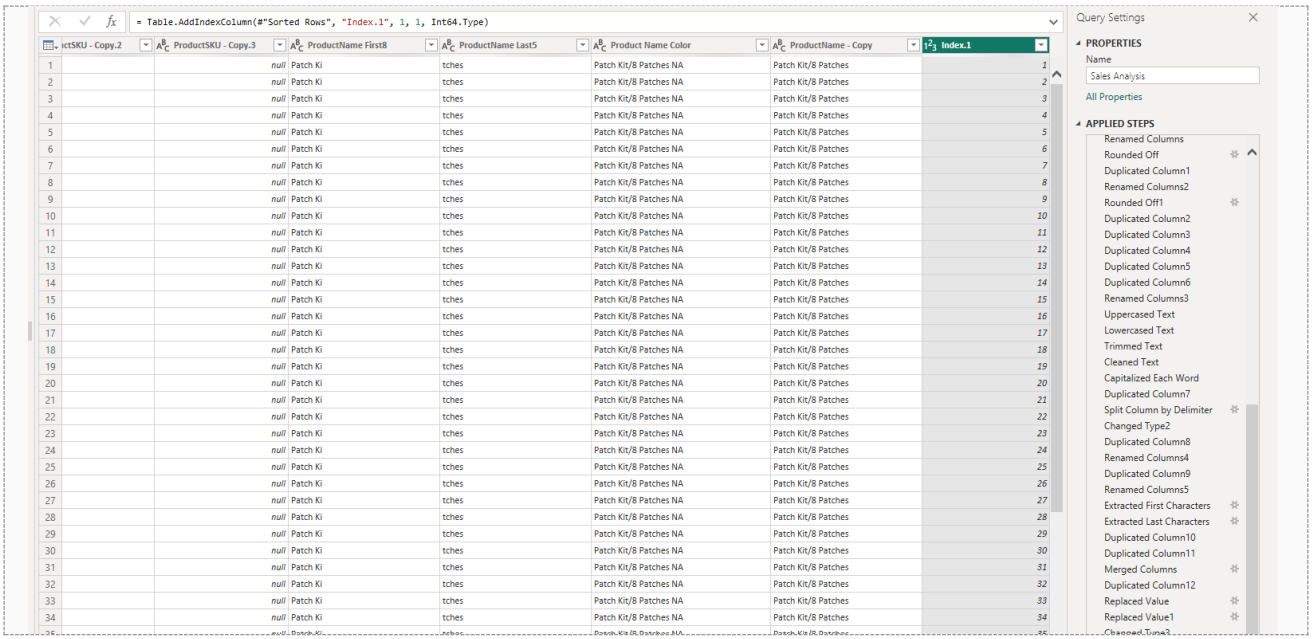


	CustomerKey	Average sales
1	14657	712.0305
2	14734	577.7266333
3	16455	28.98
4	29174	1084.8975
5	29255	880.714
6	29275	1445.969467
7	14947	1070.9961
8	18746	1178.91258
9	18906	704.65525
10	19084	1166.2825
11	22765	1198.976
12	24713	7.28
13	12483	2824.8448
14	29222	1458.203333
15	17640	24.99
16	12485	947.4797
17	29166	1087.145
18	18900	1036.555
19	18721	1777.75
20	24035	13.98
21	19083	2879.91625
22	18738	1169.3545
23	16563	19.84168
24	18769	1784.743333
25	29168	869.64684
26	29156	3578.27
27	15075	26.20806667
28	29208	3578.27
29	29207	3578.27
30	15519	21.98
31	18930	1448.195625
32	18731	1461.5942
33	16568	24.98666667

30. Create Index Column

Adds a simple unique row number, which is handy as a row identifier or surrogate key.

```
Table.AddIndexColumn(Sales Analysis, "Index", 1, 1)
```



	ictSKU - Copy.2	ProductSKU - Copy.3	ProductName First8	ProductName Last5	Product Name Color	ProductName - Copy	Index.1
1		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	1
2		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	2
3		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	3
4		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	4
5		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	5
6		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	6
7		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	7
8		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	8
9		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	9
10		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	10
11		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	11
12		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	12
13		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	13
14		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	14
15		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	15
16		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	16
17		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	17
18		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	18
19		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	19
20		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	20
21		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	21
22		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	22
23		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	23
24		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	24
25		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	25
26		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	26
27		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	27
28		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	28
29		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	29
30		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	30
31		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	31
32		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	32
33		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	33
34		null	Patch Ki	tches	Patch Kit/8 Patches NA	Patch Kit/8 Patches	34

Bonus Tasks

Sales Band (Low / Medium / High)

Buckets each transaction's value into simple tiers, easier to filter and report on than raw amounts.

```
if [Sales] <= 500 then "Low" else if [Sales] <= 2000 then "Medium" else "High"
```

Shipping Performance (Fast / Normal / Delayed)

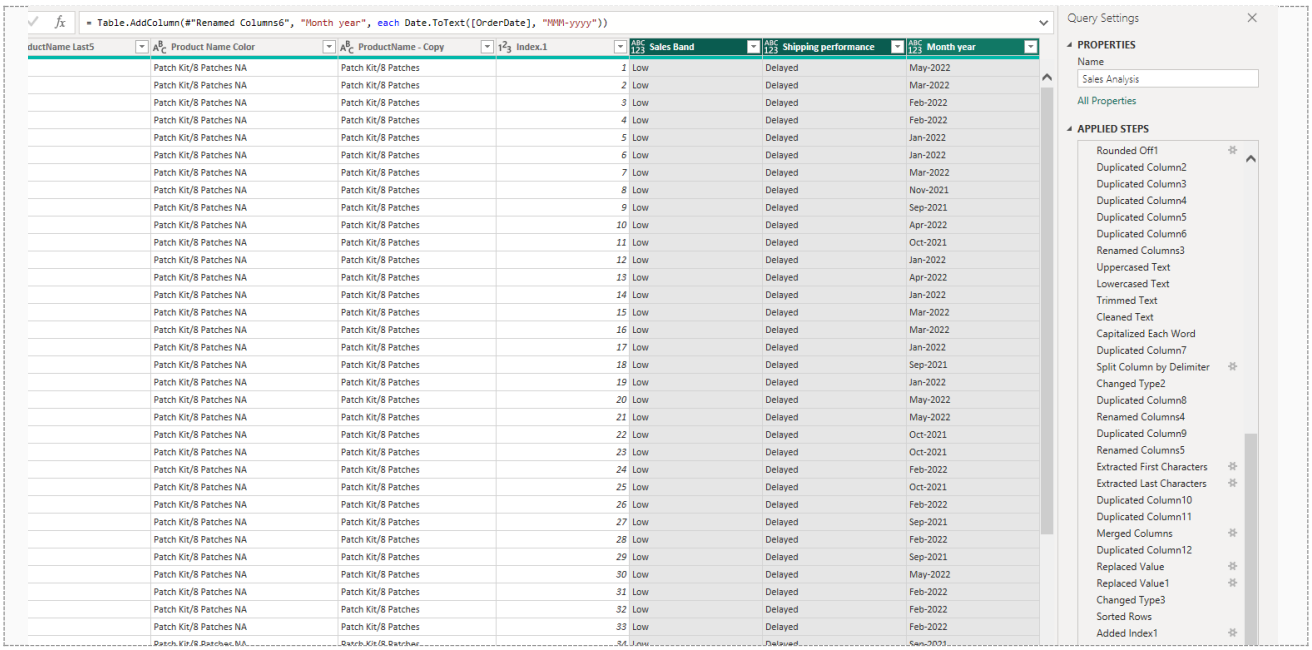
Converts the raw days-difference number into a performance label, so dispatch efficiency can be tracked at a glance.

```
if [Days Difference] <= 3 then "Fast" else if [Days Difference] <= 7 then "Normal" else "Delayed"
```

Month-Year from Order Date (e.g. Jan-2020)

Combines month and year into one sortable text label — the standard x-axis format used in time-series charts.

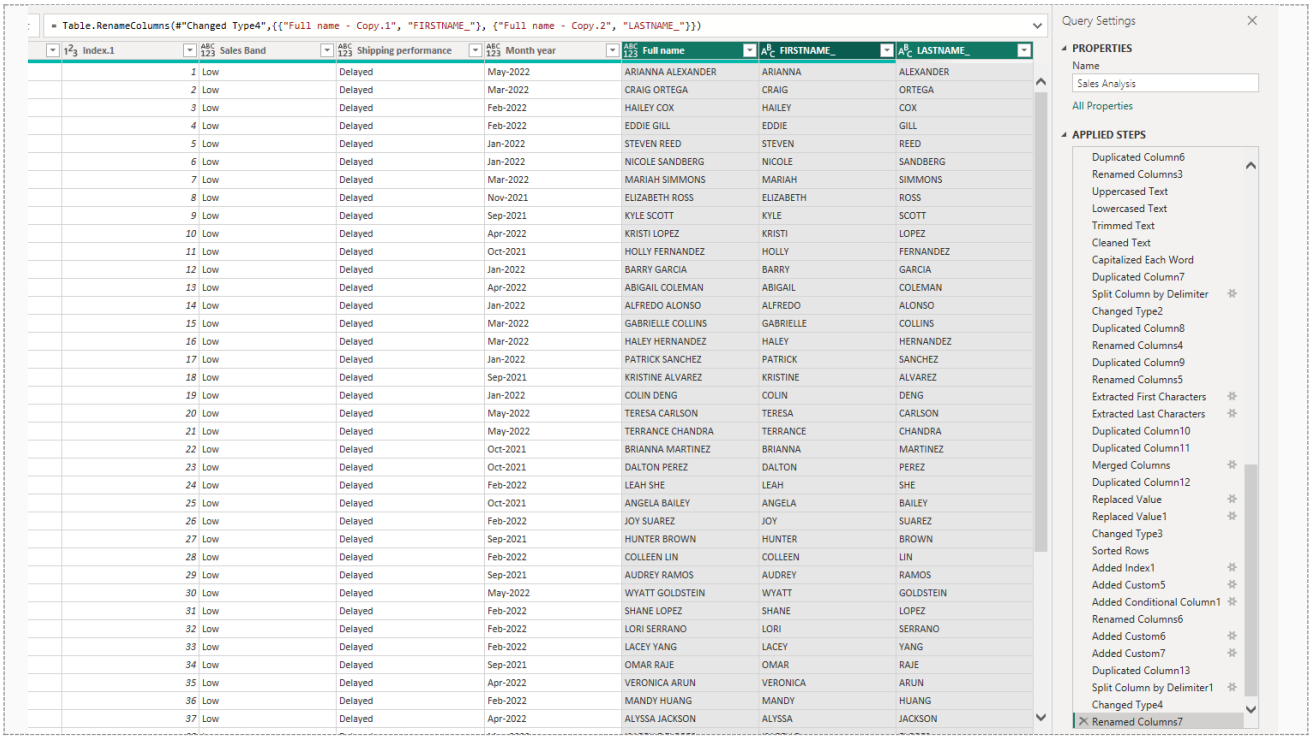
```
Date.ToText([OrderDate], "MMM-yyyy")
```



Split Customer Full Name

Demonstrates splitting one combined text field into two structured columns (first/last name), a common data-cleaning step.

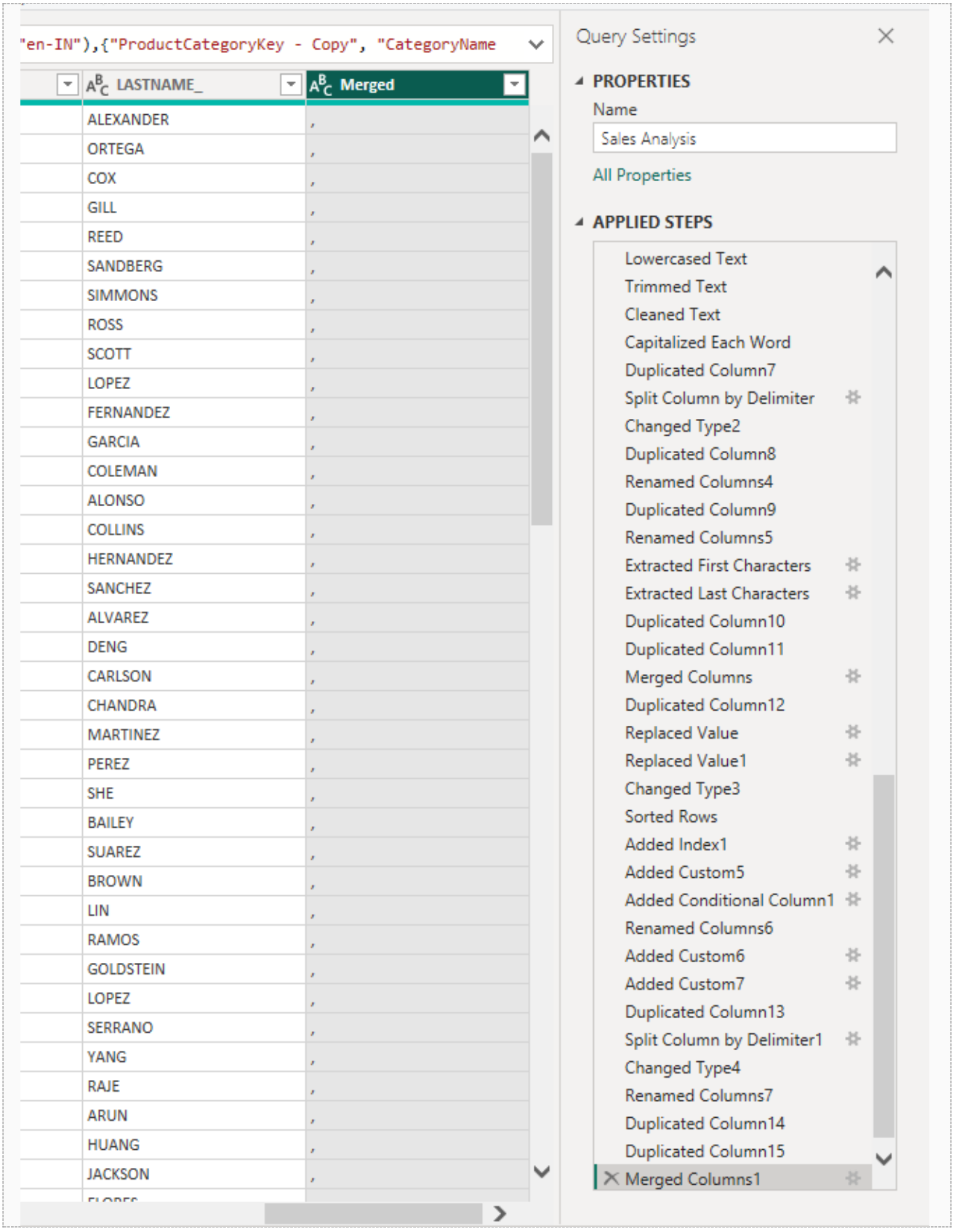
```
Full Name = [FirstName] & " " & [LastName]
then Table.SplitColumn(..., "Full Name", Splitter.SplitTextByDelimiter(" "))
```



Merge Product Category and Subcategory

Creates a single hierarchical label combining both classification levels, useful for a compact display field.

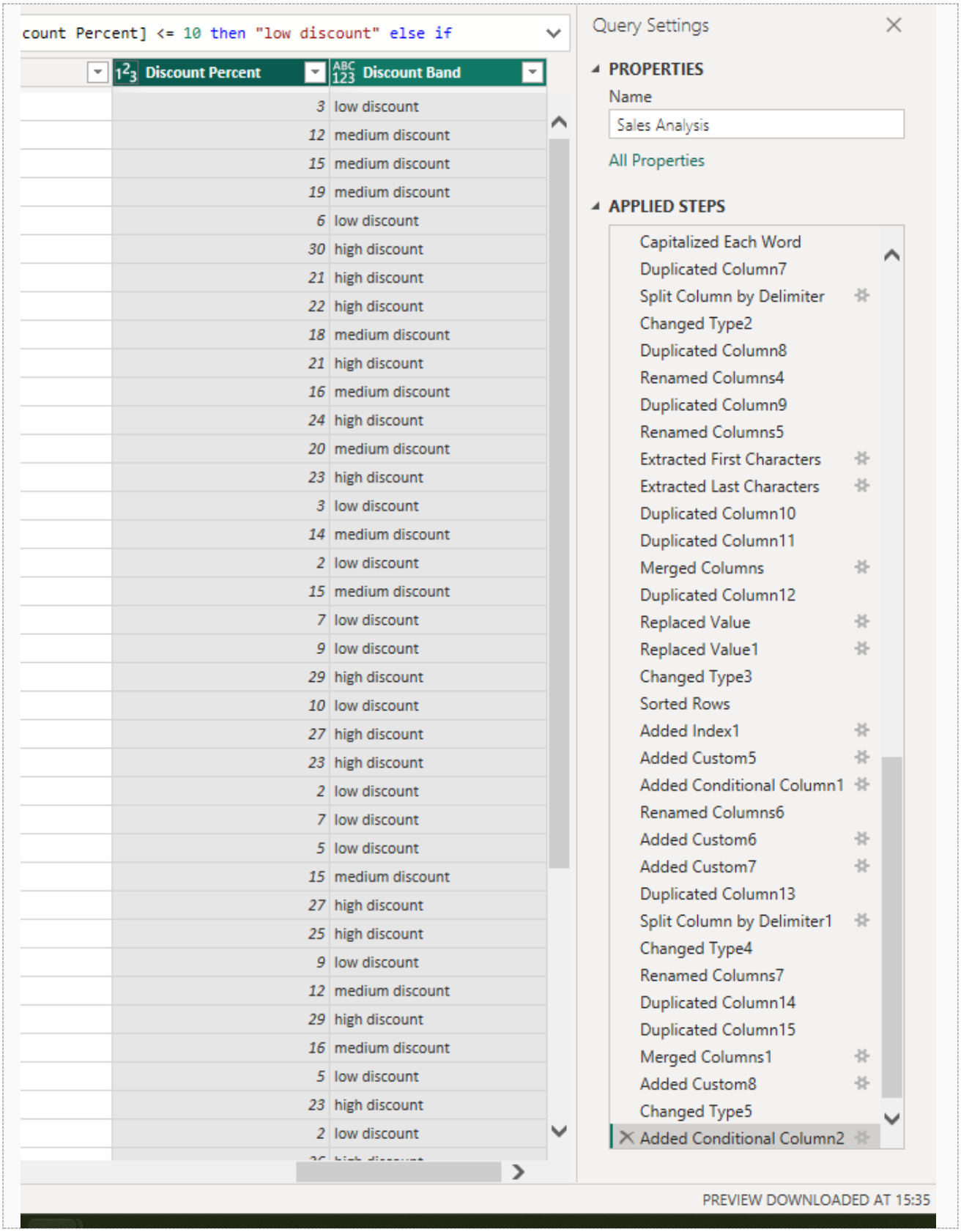
```
[CategoryName] & " - " & [SubcategoryName]
```



Discount Percentage Band

Buckets a discount percentage into tiers, useful for analysing how promotions are distributed across sales.
Note: Adventure Works has no real discount field, so this value is simulated for the exercise.

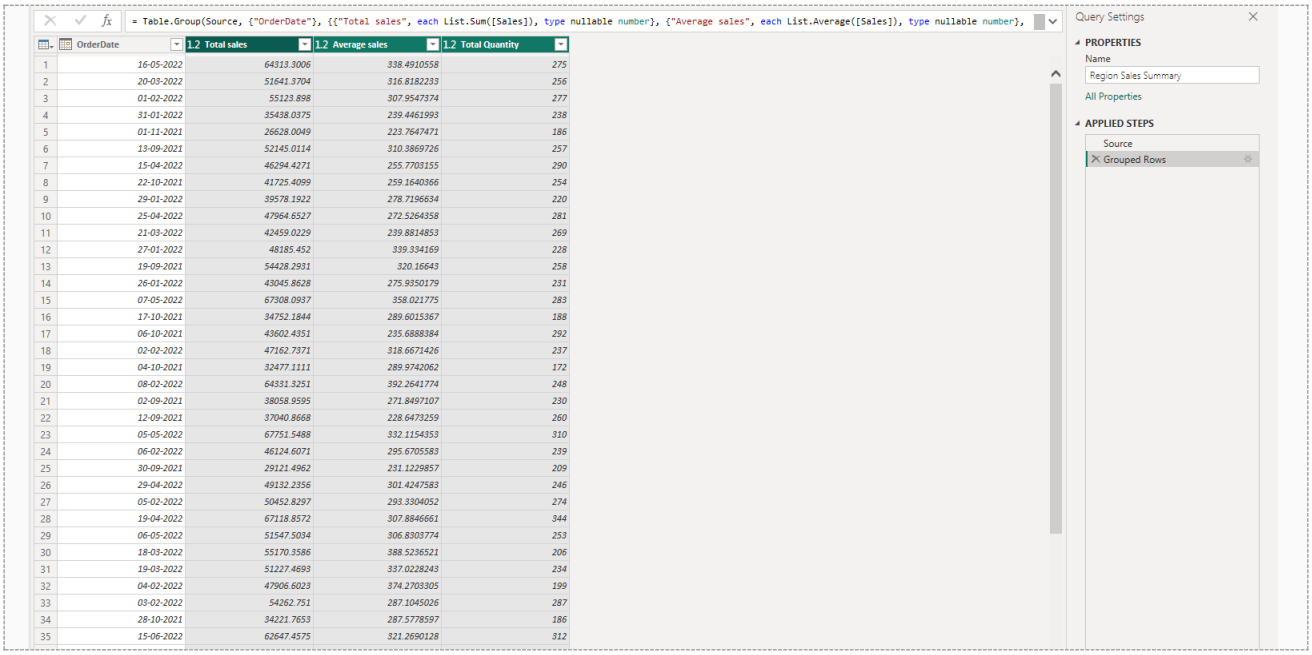
```
Discount % = Number.Round(Number.RandomBetween(0,30), 0)
then: if [Discount %] = 0 then "No Discount" else if [Discount %] <= 10 then "Low Discount"
else if [Discount %] <= 20 then "Medium Discount" else "High Discount"
```



Group by Region — Total Sales, Average Sales, Total Quantity

Rolls up transactions into one row per region with three summary metrics at once — a typical regional performance report.

```
Table.Group(Sales Analysis, {"Region"}, {{"Total Sales", each List.Sum([Sales]), type number}, {"Average Sales", each List.Average([Sales]), type number}, {"Total Quantity", each List.Sum([OrderQuantity]), Int64.Type}})
```



= Table.Group(Source, {"OrderDate"}, {{"Total sales", each List.Sum([Sales]), type nullable number}, {"Average sales", each List.Average([Sales]), type nullable number},

	OrderDate	Total sales	Average sales	Total Quantity
1	16-05-2022	64313.3006	338.4910558	275
2	20-03-2022	51641.3704	316.8182233	256
3	01-02-2022	55123.898	307.9547374	277
4	31-01-2022	35438.0375	239.4461993	238
5	01-11-2021	26628.0049	223.7647471	186
6	13-09-2021	52145.0114	310.3869726	257
7	15-04-2022	46294.4271	255.7703155	290
8	22-10-2021	41725.4099	259.1640366	254
9	29-01-2022	39578.1922	278.7196634	220
10	25-04-2022	47964.6527	272.5264358	281
11	21-03-2022	42459.0229	239.8814853	269
12	27-01-2022	48185.452	339.334169	228
13	19-09-2021	54428.2931	320.16643	258
14	26-01-2022	43045.8628	275.9350179	231
15	07-05-2022	67308.0937	358.021775	283
16	17-10-2021	34752.1844	289.6015367	188
17	06-10-2021	43602.4351	235.6888384	292
18	02-02-2022	47162.7371	318.6671426	237
19	04-10-2021	32477.1111	289.9742062	172
20	08-02-2022	64331.3251	392.2641774	248
21	02-09-2021	38058.9595	271.8497107	230
22	12-09-2021	37040.8668	228.6473259	260
23	05-05-2022	67751.5488	332.1154353	310
24	06-02-2022	46124.6071	295.6705583	239
25	30-09-2021	29121.4962	231.1229857	209
26	29-04-2022	49132.2356	301.4247583	246
27	05-02-2022	50452.8297	293.3304052	274
28	19-04-2022	67118.8572	307.8846661	344
29	06-05-2022	51547.5034	306.8303774	253
30	18-03-2022	55170.3586	388.5236521	206
31	19-03-2022	51227.4693	337.0228243	234
32	04-02-2022	47906.6023	374.2703305	199
33	03-02-2022	54262.751	287.1045026	287
34	28-10-2021	34221.7653	287.5778597	186
35	15-06-2022	62647.4575	321.2690128	312

Query Settings

PROPERTIES

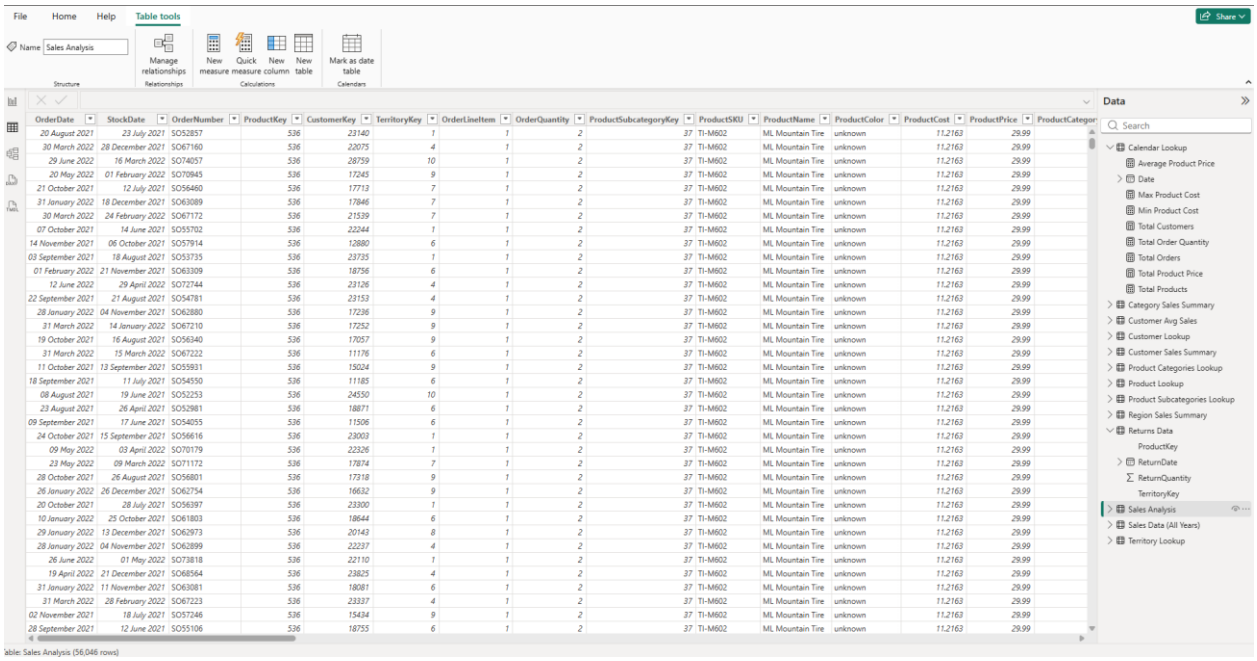
Name: Region Sales Summary

All Properties

APPLIED STEPS

Source

Grouped Rows



File Home Help Table tools

Name: Sales Analysis

Manage relationships | New measure | Quick measure | New column | New table | Mark as date table

Share

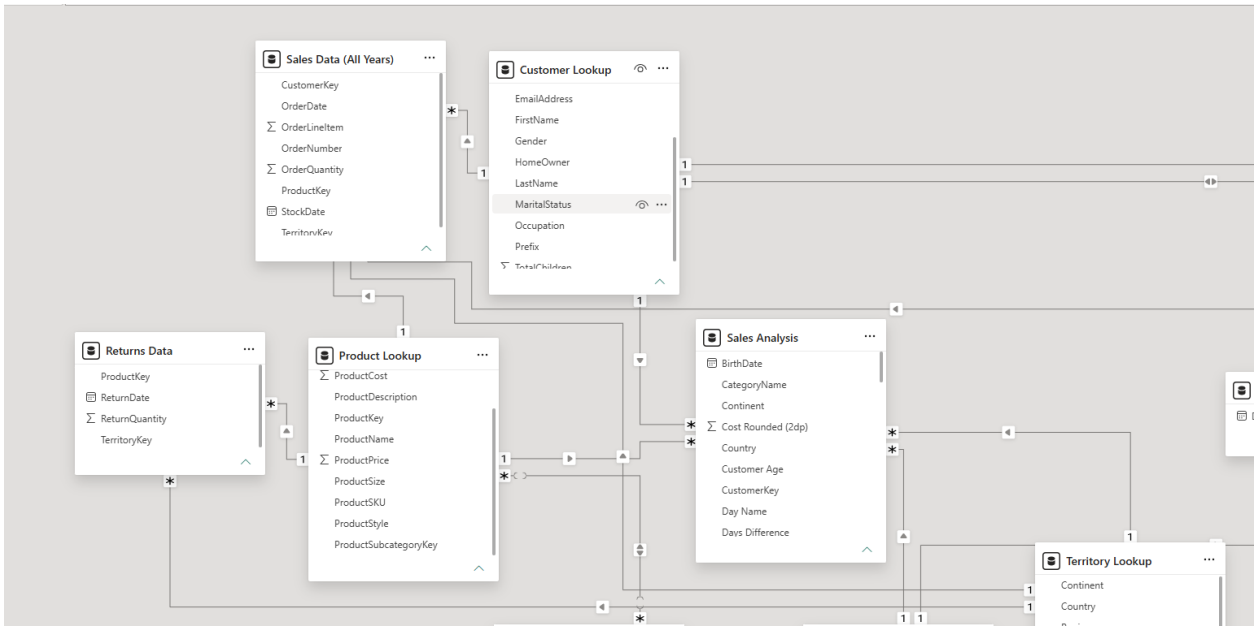
Data

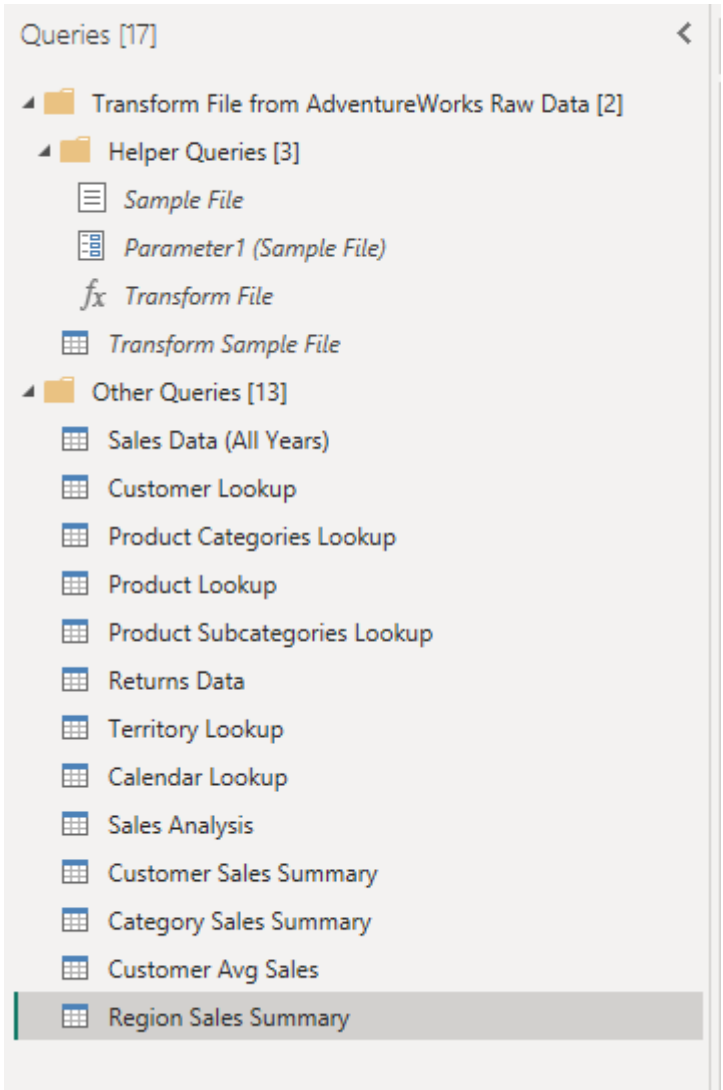
Search

Calendar Lookup
Average Product Price
Date
Max Product Cost
Min Product Cost
Total Customers
Total Order Quantity
Total Orders
Total Product Price
Total Products
Category Sales Summary
Customer Avg Sales
Customer Lookup
Customer Sales Summary
Product Categories Lookup
Product Lookup
Product Subcategories Lookup
Region Sales Summary
Returns Data
ProductKey
ReturnDate
ReturnQuantity
TerritoryKey
Sales Analysis
Sales Data (All Years)
Territory Lookup

OrderDate	StockDate	OrderNumber	ProductKey	CustomerKey	TerritoryKey	OrderLineItem	OrderQuantity	ProductSubcategoryKey	ProductSKU	ProductName	ProductColor	ProductCost	ProductPrice	ProductCategor
20 August 2021	23 July 2021	SO52857	536	23140	1	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
30 March 2022	28 December 2021	SO67160	536	23075	4	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
29 June 2022	16 March 2022	SO74057	536	28759	10	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
20 May 2022	01 February 2022	SO70945	536	17245	9	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
21 October 2021	12 July 2021	SO56460	536	17713	7	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
31 January 2022	18 December 2021	SO63089	536	17846	7	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
30 March 2022	24 February 2022	SO67172	536	21539	7	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
07 October 2021	14 June 2021	SO55702	536	22244	1	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
14 November 2021	06 October 2021	SO57914	536	12880	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
03 September 2021	18 August 2021	SO53735	536	23735	1	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
01 February 2022	21 November 2021	SO63309	536	18756	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
12 June 2022	29 April 2022	SO72744	536	23126	4	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
22 September 2021	21 August 2021	SO54781	536	23153	4	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
28 January 2022	04 November 2021	SO62880	536	17236	9	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
31 March 2022	14 January 2022	SO67210	536	17252	9	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
19 October 2021	16 August 2021	SO56340	536	17057	9	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
31 March 2022	15 March 2022	SO67222	536	11176	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
11 October 2021	13 September 2021	SO55931	536	15024	9	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
18 September 2021	11 July 2021	SO54550	536	11185	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
08 August 2021	19 June 2021	SO52253	536	24550	10	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
23 August 2021	26 April 2021	SO52981	536	18871	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
09 September 2021	17 June 2021	SO54055	536	11506	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
24 October 2021	15 September 2021	SO56616	536	23003	1	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
09 May 2022	03 April 2022	SO70179	536	22326	1	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
23 May 2022	09 March 2022	SO71172	536	17874	7	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
28 October 2021	26 August 2021	SO56801	536	17318	9	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
26 January 2022	26 December 2021	SO62754	536	16632	9	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
20 October 2021	28 July 2021	SO56397	536	23300	1	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
10 January 2022	25 October 2021	SO61803	536	18644	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
29 January 2022	13 December 2021	SO62973	536	20143	8	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
28 January 2022	04 November 2021	SO62899	536	22237	4	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
26 June 2022	01 May 2022	SO73818	536	22110	1	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
19 April 2022	21 December 2021	SO68564	536	23825	4	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
31 January 2022	11 November 2021	SO63081	536	18081	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
31 March 2022	28 February 2022	SO67223	536	23337	4	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
02 November 2021	18 July 2021	SO57246	536	15434	9	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	
28 September 2021	12 June 2021	SO55106	536	18755	6	1	2	37	TI-M602	ML Mountain Tire	unknown	11.2163	29.99	

Table: Sales Analysis (56,046 rows)





Conclusion

Through this practical, I got a good hands-on understanding of how Power BI's built-in aggregations and Power Query editor work together — from simple SUM/AVERAGE measures to more involved tasks like merging tables, handling messy real-world data (like the broken BirthDate rows I encountered), building

conditional columns, and grouping data for summary insights. It also helped me understand the difference between a relationship-based data model and a fully merged/flattened working table, and when each approach is more useful.